

LimberCloud: Scalable Analytic Computation of Limber Power Spectra for Stage-IV Cosmology

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Abstract. Next-generation imaging surveys demand rapid and accurate evaluation of angular power spectra across high-dimensional parameter spaces. We present LimberCloud, a tensorised reformulation of the Limber integral that algebraically separates cosmology-dependent quantities from survey-specific tomographic redshift distributions. The factorisation reduces each line-of-sight integral to closed-form expressions, eliminating numerical quadrature entirely. Validated against the Core Cosmology Library for the LSST Y1 3×2 pt data vector, the method achieves sub-per-cent accuracy across all tomographic bin pairs and multipoles $20 \leq \ell \leq 2000$. A GPU-accelerated JAX implementation evaluates the full set of angular power spectra in approximately 3 ms—over three orders of magnitude faster than conventional approaches—and the cosmology-dependent coefficients are structured for end-to-end emulation, offering a further three-to-four order-of-magnitude speed-up at inference time. The algebraic structure also reveals that the predicted spectra depend on the redshift distribution uncertainties as an exact low-order polynomial, enabling analytic marginalisation that avoids the finite-difference and fixed-cosmology approximations of existing treatments. Extensions to non-flat geometries, CMB lensing, and beyond-Limber scales are outlined.

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1 Introduction

Observations of the large-scale structure of the Universe, combining weak gravitational lensing with galaxy clustering, provide a powerful probe of cosmology [1–4]. Over the past decade, wide-field imaging surveys such as the Kilo-Degree Survey [KiDS; 5, 6], the Dark Energy Survey [DES; 7, 8], and the Hyper Suprime-Cam Survey [HSC; 9, 10] have delivered increasingly precise measurements of the auto- and cross-correlations of galaxy shapes and positions [11–16, 83, 84]. These two-point functions form the foundation of the now-standard 3×2 pt framework [18–22], enabling stringent constraints on late-time structure growth and the cosmic expansion history at a level of precision increasingly competitive with early-Universe measurements from the cosmic microwave background [CMB; 23].

The next generation of Stage-IV facilities—the Vera C. Rubin Observatory [Rubin¹; 24], *Euclid*² [25], the Nancy Grace Roman Space Telescope [Roman³; 26], and the China Space

¹<https://rubinobservatory.org>

²<https://www.cosmos.esa.int/web/euclid>

³<https://roman.gsfc.nasa.gov>

Station Telescope [CSST⁴; 27]—will extend sky coverage, depth, and image quality by orders of magnitude. A central ingredient of the resulting analyses is the computation of angular power spectra under the Limber approximation [29, 30], which underpins not only the 3×2 pt framework but any inference based on projected anisotropies of the matter and tracer fields [28]. These projections must be evaluated repeatedly—often millions of times [e.g. 78]—across parameter spaces that span cosmology, astrophysical modelling, and survey-level systematics, placing severe demands on theoretical pipelines.

In current analyses, the cost of Limber integrals grows rapidly with multipole resolution, the number of tomographic bins, and the complexity of the nuisance-parameter model. Modern pipelines incorporate detailed baryonic-feedback prescriptions [e.g. 33–37], scale-dependent galaxy-bias models [e.g. 38–40], flexible intrinsic-alignment models [e.g. 43–48], and sophisticated treatments of tomographic redshift distribution uncertainties [e.g. 49–54]. Each extension enlarges the parameter space, and the forward-modelling step—recomputing angular power spectra at every likelihood evaluation—often dominates the total inference cost.

A range of strategies have been developed to alleviate this bottleneck. Emulators of the non-linear matter power spectrum [57–63], together with recent factorisation-based decompositions of the matter power spectrum into a small set of cosmology-independent basis functions [91], accelerate the three-dimensional modelling stage but do not directly emulate angular power spectra, which depend on survey-specific selection functions and tomographic redshift distributions. GPU-enabled cosmology frameworks based on JAX [64] and Julia [66]—including JAX-Cosmo [67] and LimberJack.jl [68]—accelerate the line-of-sight projection itself, and additionally enable gradient-based samplers that reduce the number of likelihood evaluations required. Complementary approaches reformulate the Limber and beyond-Limber integrals using FFTLog [86] expansions or specialised basis functions to handle oscillatory kernels efficiently [e.g. 69–75]; recent implementations such as Blast.jl [76] and Swift C_ℓ [77] combine these techniques with modern hardware to achieve significant runtime reductions. At the sampler level, frameworks such as Cobaya [80] and CosmoSIS [81] exploit the observation that nuisance parameters typically do not affect the three-dimensional power spectrum, enabling a fast–slow decomposition in which the expensive cosmological computation is reused across multiple nuisance-parameter proposals. However, all of these approaches intertwine cosmology-dependent quantities with survey-specific components such as tomographic binning and tomographic redshift distributions, so the Limber projection must still be re-evaluated whenever any nuisance parameter—in particular the tomographic redshift distributions—changes, even when the underlying cosmology is held fixed.

A formulation that cleanly separates these two classes of dependence would enable direct emulation of the cosmological projection coefficients and open the door to analytic marginalisation over nuisance parameters, substantially reducing the burden on high-dimensional sampling [49, 52, 53, 79, 85]. In this work, we present LimberCloud, a tensorised analytic reformulation of the Limber projection that achieves precisely this separation. By piecewise-linearising the radial kernels and the matter power spectrum on a discrete redshift grid, the standard Limber integrals reduce to compact tensor contractions of the form $C_\ell^{ab} = \sum_{ij} B_\ell^{ij} \phi_i^a \phi_j^b$, where the coefficients B_ℓ^{ij} depend only on cosmology and the tomographic redshift distribution vectors ϕ_i^a and ϕ_j^b encode all survey-specific information. This factorisation is exact in the limit of infinitely fine redshift binning and achieves sub-per-cent

⁴<http://www.bao.ac.cn/csst/>

agreement with the Core Cosmology Library [CCL; 55] for the bin widths adopted in current Stage-IV forecasts. Implemented in JAX, the framework exploits just-in-time compilation, automatic vectorisation, and GPU acceleration; in the large-batch regime relevant for inference, the JAX–GPU implementation outperforms CCL by more than an order of magnitude. The tensorised structure further provides a natural foundation for future end-to-end emulation of the cosmology-dependent coefficients and for analytic marginalisation over nuisance parameters, both of which we leave to follow-up work. The present work focuses on the mathematical formulation, its validation against CCL, and computational benchmarks across CPU and GPU architectures.

The remainder of this paper is organised as follows. Section 2 reviews the theoretical formalism of angular power spectra under the Limber approximation. Section 3 describes the validation setup. Section 4 introduces the linearisation of the kernel functions and matter power spectra. Section 5 presents the tensorised reformulation of the Limber projection and compares its accuracy with benchmark calculations. Section 6 evaluates performance on CPU and GPU architectures. Section 7 summarises the framework, discusses analytic marginalisation prospects, and outlines future directions.

2 Theoretical Formalism

The standard 3×2 pt framework combines three complementary two-point statistics constructed from foreground lens galaxies, for which precise distance estimates are available, and background source galaxies with robust shape measurements. These include: (i) the auto-correlation of observed galaxy ellipticities ($\epsilon\epsilon$), commonly referred to as cosmic shear; (ii) the angular auto-correlation of the lens-galaxy positions ($\theta\theta$), hereafter galaxy clustering; and (iii) the cross-correlation between lens positions and source ellipticities ($\theta\epsilon$), known as galaxy–galaxy lensing.

Each observable receives cosmological contributions from weak gravitational lensing, denoted γ , and from the large-scale structure traced by galaxies, denoted g . In addition, astrophysical systematics contribute to the measured correlations, most notably intrinsic alignments of galaxy shapes, I , and cosmic magnification, μ . The present analysis neglects redshift-space distortions (RSD) and does not model CMB secondary anisotropies—including CMB lensing, the thermal and kinetic Sunyaev–Zel’dovich effects, and the Integrated Sachs–Wolfe (ISW) effect—although these follow the same projected-kernel formalism and can be incorporated straightforwardly in future work.

Under the Limber approximation, the angular power spectrum between any two projected tracers (u, v), defined in tomographic redshift bins (a, b), can be written as

$$C_{uv}^{ab}(\ell) = A_{uv}(\ell) \int_0^{\chi_H} d\chi \frac{W_u^a(\chi) W_v^b(\chi)}{f_K^2(\chi)} P_{uv} \left[k = \frac{\ell + 1/2}{f_K(\chi)}, z(\chi) \right], \quad (2.1)$$

where the comoving ^{angular diameter} transverse distance $f_K(\chi)$ is defined as

$$f_K(\chi) = \begin{cases} K^{-1/2} \sin(K^{1/2} \chi), & K > 0, \\ \chi, & K = 0, \\ |K|^{-1/2} \sinh(|K|^{1/2} \chi), & K < 0, \end{cases} \quad (2.2)$$

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with $K \equiv \Omega_K H_0^2 / c^2$ denoting the spatial curvature. This distance enters both the geometric lensing kernel and the mapping between angular multipole ℓ and the three-dimensional wavenumber k .

The prefactor $A_{uv}(\ell)$ is a multipole-dependent correction that accounts for the spin of the projected fields; it arises from the exact angular power spectrum expression and reduces to unity only for scalar–scalar correlations. Its explicit form for each tracer pair is given in Table 1. The quantity $P_{uv}(k, z)$ is the corresponding three-dimensional power spectrum, while $W_u^a(\chi)$ and $W_v^b(\chi)$ are the radial projection kernels associated with tracers u and v respectively, whose functional forms—either the number-density kernel W_ϕ or the lensing convergence kernel W_κ —are specified for each tracer pair in Table 1.

In this work, we adopt a unified treatment of all tracer kernels. All tracer-specific physics—galaxy bias, intrinsic-alignment amplitudes, and magnification response—is absorbed entirely into multiplicative prefactors within the three-dimensional power spectrum $P_{uv}(k, z)$. As a result, all projected observables are constructed from the same pair of kernel functions, corresponding to galaxy number density and gravitational lensing convergence. This formulation is compact, fully general, and readily extensible to additional tracers or physical effects.

The clustering, or galaxy number-density, kernel is given directly by the tomographic redshift distribution of the tracer population,

$$W_\phi^a(\chi) \equiv \phi^a(\chi) = \phi^a(z) \frac{H(z)}{c}, \quad (2.3)$$

where $\phi^a(z)$ denotes the tomographic redshift distribution of galaxies in bin a , and the equality follows from $d\chi = c dz / H(z)$. This kernel specifies the radial distribution of galaxies in bin a and determines how they trace the underlying matter field along the line of sight.

The kernel associated with weak gravitational lensing convergence is

$$W_\kappa^a(\chi) = \frac{3\Omega_m H_0^2}{2c^2} \frac{f_K(\chi)}{a(\chi)} \int_\chi^{\chi_H} d\chi' \phi^a(\chi') \frac{f_K(\chi' - \chi)}{f_K(\chi')}. \quad (2.4)$$

This lensing kernel quantifies the efficiency with which matter at comoving distance χ lenses background galaxies, encapsulating both the geometric lensing strength and its dependence on the cosmic expansion history.

The observed galaxy ellipticity field, ϵ , receives contributions from gravitational shear, γ , and from intrinsic alignments, $\mathbf{I}$. Cosmic shear—the auto-correlation of observed ellipticities—therefore probes the integrated tidal field along the line of sight. The tomographic shape–shape power spectra decompose as

$$C_{\epsilon\epsilon}^{ab}(\ell) = C_{\gamma\gamma}^{ab}(\ell) + C_{\gamma\mathbf{I}}^{ab}(\ell) + C_{\mathbf{I}\gamma}^{ab}(\ell) + C_{\mathbf{I}\mathbf{I}}^{ab}(\ell), \quad (2.5)$$

where the indices a and b label the tomographic source bins.

Galaxy–galaxy lensing probes the correlation between the projected number density of foreground lens galaxies and the shapes of background source galaxies, providing a direct link between the galaxy distribution and the underlying matter field. The observed position–shape correlations receive contributions from gravitational lensing, intrinsic alignments, and cosmic magnification of the foreground sample. The tomographic number–shape angular power spectra therefore decompose as

$$C_{\theta\epsilon}^{ab}(\ell) = C_{g\gamma}^{ab}(\ell) + C_{g\mathbf{I}}^{ab}(\ell) + C_{\mu\gamma}^{ab}(\ell) + C_{\mu\mathbf{I}}^{ab}(\ell), \quad (2.6)$$

where a labels the lens bin and b the source bin.

Galaxy clustering probes the projected spatial distribution of foreground lens galaxies and traces the underlying large-scale structure. In addition to intrinsic clustering driven by matter fluctuations, the observed galaxy number density is affected by lensing magnification, which modifies the apparent density through flux and size biases. The observed tomographic position–position angular power spectra therefore decompose as

$$C_{\theta\theta}^{ab}(\ell) = C_{\text{gg}}^{ab}(\ell) + C_{\text{g}\mu}^{ab}(\ell) + C_{\mu\text{g}}^{ab}(\ell) + C_{\mu\mu}^{ab}(\ell), \quad (2.7)$$

where a and b both label lens bins.

Table 1 summarises the prefactor, kernel combination, and three-dimensional power spectrum for each of the twelve contributions. All individual angular power spectra are obtained by evaluating Equation 2.1 with the entries specified in the corresponding row.

We model three classes of astrophysical effects through bias prefactors in the three-dimensional power spectra: galaxy bias $b_g(k, z)$, intrinsic-alignment bias $b_I(k, z)$, and the magnification response $b_\mu(z)$. The galaxy and intrinsic-alignment biases are allowed to depend on both scale and redshift, while the magnification response is determined solely by the slope of the galaxy number counts and is therefore taken to be scale independent.

3 Fiducial Survey Configuration

This section specifies the fiducial model, astrophysical parameters, tomographic redshift distributions, and reference data vectors used throughout the validation. The configuration follows the LSST DESC Science Requirements Document [SRD; 24] for a Y1 survey scenario with five lens and five source tomographic bins. We have verified that all results generalise to the Y10 survey scenario with ten lens bins; those results are not shown here for brevity.

3.1 Fiducial cosmology

The fiducial cosmology is a spatially flat Λ CDM model consistent with the *Planck* 2018 constraints [23]. The primary parameters are: $\Omega_m = 0.3152$, $\Omega_b = 0.0493$, $\Omega_{\text{CDM}} = 0.2645$, $h = 0.6736$, $n_s = 0.9649$, $A_s = 2.083 \times 10^{-9}$, and $\Omega_K = 0$. The dark-energy equation of state is fixed to the cosmological-constant values $w_0 = -1$ and $w_a = 0$. Neutrino masses are treated in a single-species approximation with $\sum m_\nu = 0.06$ eV and $N_{\text{eff}} = 3.033$. Linear transfer functions are computed with CAMB [87], and the non-linear matter power spectrum is obtained from the HMCode2020 prescription [35] with baryonic feedback parameterised by $\log_{10}(T_{\text{AGN}}/\text{K}) = 7.8$.

3.2 Astrophysical parameters

The three classes of astrophysical bias defined in Table 1 are specified as follows. The linear galaxy bias in the lens sample is modelled as

$$b_g(z) = \frac{b_*}{D(z)}, \quad (3.1)$$

where $D(z)$ is the linear growth factor normalised to unity at $z = 0$ and $b_* = 1.05$ for the Y1 survey scenario [38]. This parameterisation ensures that the effective bias increases with redshift in a manner consistent with the evolution of dark-matter halo abundances, following conventions adopted in LSST DESC forecasts.

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Table 1. Summary of the 3×2 pt angular power spectra considered in this work. All observables are computed under the Limber approximation using a unified pair of radial kernels: the number-density kernel W_ϕ and the lensing convergence kernel W_κ . Multipole-dependent prefactors $A_{uv}(\ell)$ account for the spin of the projected fields.

Observable	Contribution	$A_{uv}(\ell)$	Projected kernels	3D power spectrum
$C_{\epsilon\epsilon}^{ab}$	$\gamma\gamma$	$\frac{(\ell+2)!}{(\ell-2)!} \frac{1}{(\ell+1/2)^4}$	$W_\kappa^a W_\kappa^b$	$P_{\delta\delta}$
	γI	$\frac{(\ell+2)!}{(\ell-2)!} \frac{1}{(\ell+1/2)^4}$	$W_\kappa^a W_\phi^b$	$b_{\text{I}}(k, z) P_{\delta\delta}$
	$\text{I}\gamma$	$\frac{(\ell+2)!}{(\ell-2)!} \frac{1}{(\ell+1/2)^4}$	$W_\phi^a W_\kappa^b$	$b_{\text{I}}(k, z) P_{\delta\delta}$
	II	$\frac{(\ell+2)!}{(\ell-2)!} \frac{1}{(\ell+1/2)^4}$	$W_\phi^a W_\phi^b$	$b_{\text{I}}^2(k, z) P_{\delta\delta}$
$C_{\theta\epsilon}^{ab}$	$\text{g}\gamma$	$\sqrt{\frac{(\ell+2)!}{(\ell-2)!}} \frac{1}{(\ell+1/2)^2}$	$W_\phi^a W_\kappa^b$	$b_{\text{g}}(k, z) P_{\delta\delta}$
	gI	$\sqrt{\frac{(\ell+2)!}{(\ell-2)!}} \frac{1}{(\ell+1/2)^2}$	$W_\phi^a W_\phi^b$	$b_{\text{g}}(k, z) b_{\text{I}}(k, z) P_{\delta\delta}$
	$\mu\gamma$	$\sqrt{\frac{(\ell+2)!}{(\ell-2)!}} \frac{\ell(\ell+1)}{(\ell+1/2)^4}$	$W_\kappa^a W_\kappa^b$	$b_\mu(z) P_{\delta\delta}$
	μI	$\sqrt{\frac{(\ell+2)!}{(\ell-2)!}} \frac{\ell(\ell+1)}{(\ell+1/2)^4}$	$W_\kappa^a W_\phi^b$	$b_\mu(z) b_{\text{I}}(k, z) P_{\delta\delta}$
$C_{\theta\theta}^{ab}$	gg	1	$W_\phi^a W_\phi^b$	$b_{\text{g}}^2(k, z) P_{\delta\delta}$
	$\text{g}\mu$	$\frac{\ell(\ell+1)}{(\ell+1/2)^2}$	$W_\phi^a W_\kappa^b$	$b_{\text{g}}(k, z) b_\mu(z) P_{\delta\delta}$
	μg	$\frac{\ell(\ell+1)}{(\ell+1/2)^2}$	$W_\kappa^a W_\phi^b$	$b_{\text{g}}(k, z) b_\mu(z) P_{\delta\delta}$
	$\mu\mu$	$\frac{\ell^2(\ell+1)^2}{(\ell+1/2)^4}$	$W_\kappa^a W_\kappa^b$	$b_\mu^2(z) P_{\delta\delta}$

Galaxy shape alignments are described by the non-linear alignment (NLA) model [45, 46]. The alignment bias takes the form

$$b_{\text{I}}(z) = -\frac{C_1 \bar{\rho}_{\text{m}}}{D(z)} A_{\text{IA}} \left(\frac{1+z}{1+z_{\text{pivot}}} \right)^{\eta_{\text{A}}}, \quad (3.2)$$

where $C_1 = 5 \times 10^{-14} h^{-2} M_\odot^{-1} \text{Mpc}^3$ is the normalisation constant, $\bar{\rho}_{\text{m}}$ the comoving mean matter density, and $D(z)$ the linear growth factor. The fiducial values are $A_{\text{IA}} = 0.5$, $\eta_{\text{A}} = 0$, and $z_{\text{pivot}} = 0.5$, applied uniformly across all source bins.

Lensing magnification is characterised by a per-bin constant response $b_\mu^{(i)}$, determined by the slope of the cumulative galaxy number counts at the survey flux limit. For the five lens bins in the Y1 survey scenario, the fiducial values are

$$b_\mu^{(i)} = \{0.659, 0.775, 0.815, 0.846, 0.994\}. \quad (3.3)$$

All astrophysical bias parameters are held fixed throughout the validation and are not varied in the accuracy benchmarks. Figure 1 illustrates the resulting tomographic redshift distributions

for both the lens and source samples adopted in this work; the corresponding astrophysical bias values listed above are applied to these distributions when constructing the projected angular power spectra.

3.3 Tomographic redshift distributions

Tomographic redshift distributions are constructed following the LSST DESC SRD specifications using the `binny` package [65], which implements the underlying parametric forms and binning conventions adopted in the public `forecasting` analysis-choices repository. The underlying galaxy redshift distribution is described by a Smail-type parametric form,

$$p(z) \propto \left(\frac{z}{z_*}\right)^\beta \exp\left[-\left(\frac{z}{z_*}\right)^\alpha\right], \quad (3.4)$$

with shape parameters $(\beta, \alpha, z_*) = (2.0, 0.94, 0.26)$ for the lens sample and $(2.0, 0.78, 0.13)$ for the source sample. Photometric-redshift uncertainties are incorporated by convolving this distribution with a Gaussian error model characterised by a scatter of $\sigma_z = 0.03$ for lens galaxies and $\sigma_z = 0.05$ for source galaxies, with zero bias in both cases.

The convolved distribution is divided into N_B tomographic bins per tracer sample as follows. In the Y1 survey scenario, $N_B = 5$ for both samples. Lens galaxies are assigned to five bins of equal redshift width 0.2 over the range $0.2 \leq z \leq 1.2$. Source galaxies are partitioned into five bins of equal galaxy count. The resulting tomographic redshift distributions for both samples are shown in Figure 1. Each distribution is re-sampled onto a uniform redshift grid of $N_Z + 1 = 351$ points spanning $0 \leq z \leq 3.5$ with a constant spacing of $\Delta z = 0.01$, corresponding to $N_Z = 350$ redshift intervals. All distributions are normalised to unit integral. This grid defines the resolution at which the piecewise linearisation of physical quantities is performed in Section 4.

The per-bin effective number densities, as adopted in the covariance calculation, are

$$n_{\text{eff}}^{\text{lens}} = \{0.087, 0.096, 0.140, 0.181, 0.167\} \text{ arcmin}^{-2} \quad (3.5)$$

for the five lens bins and

$$n_{\text{eff}}^{\text{source}} = \{2.090, 2.298, 2.105, 1.901, 1.839\} \text{ arcmin}^{-2} \quad (3.6)$$

for the five source bins.

The lens densities listed in Equation 3.5 are deliberately lower than the SRD baseline of approximately 18 arcmin^{-2} for the Y1 sample—roughly 3.6 arcmin^{-2} per tomographic bin when divided evenly across the five lens bins—which counts all galaxies detected above the survey flux limit. A realistic 3×2 pt lens sample requires additional cuts on photometric-redshift quality, magnification response, and colour stability that reduce the usable density by an order of magnitude. We therefore adopt per-bin densities consistent with a DES-Y6-style lens selection [93] forward-modelled to LSST in Zhang et al. [94]. This bright-population definition is essential not only for representing realistic shot noise, but also for maintaining SRD-level photometric-redshift uncertainties on the lens redshift distributions: pushing to a fainter lens sample would dramatically inflate the photo- z calibration error budget and dominate over both the statistical and modelling uncertainties, providing a less realistic uncertainty baseline against which to validate the computational technique.

Source-sample densities, by contrast, are essentially unchanged from the SRD baseline, because source selection is dominated by shape-measurement quality cuts that do not differ

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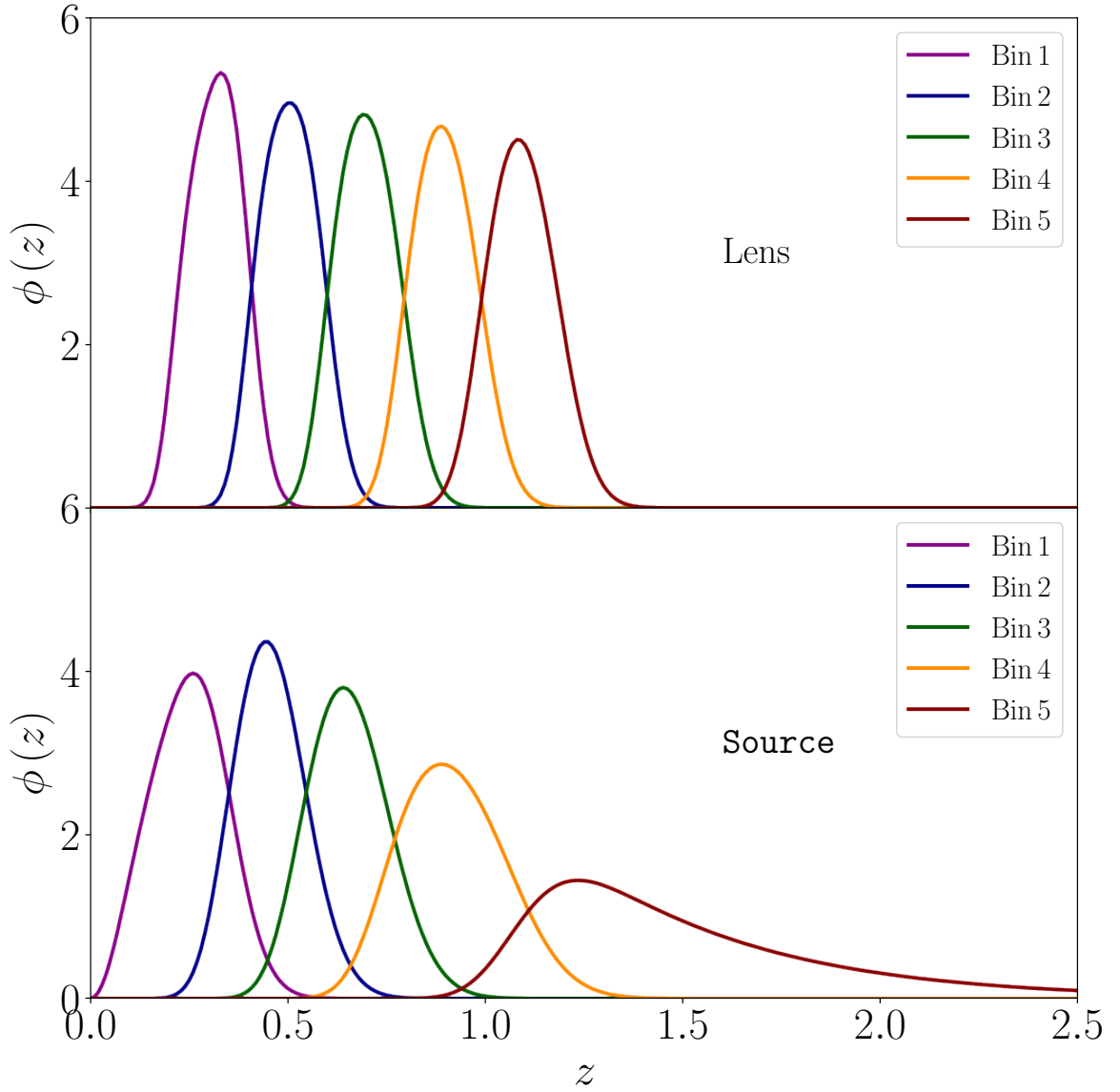


Figure 1. Tomographic redshift distributions for the LSST Y1 survey scenario. The upper panel shows the five equal-width lens bins over $0.2 \leq z \leq 1.2$; the lower panel shows the five equal-count source bins. All distributions are normalised to unit integral.

substantially between the gold-detected and analysis-quality samples. We emphasise that this choice affects only the per-bin Gaussian covariance reference curves overlaid in Figures 5–7; the intrinsic accuracy of LimberCloud assessed in Section 5 is independent of n_{eff} .

The per-component ellipticity dispersion is $\sigma_\epsilon = 0.26$ in each source bin. The survey footprint covers $18\,000 \text{ deg}^2$, corresponding to a sky fraction of $f_{\text{sky}} \simeq 0.44$. Uncertainties in the tomographic redshift distributions are not varied in the present work; their treatment via analytic marginalisation is deferred to follow-up work.

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3.4 Angular power spectra and covariance matrices

Reference angular power spectra serving as the validation benchmark are computed with PyCCL using the Limber approximation. The data vectors are evaluated at 20 logarithmically spaced multipole bins between $\ell = 20$ and $\ell = 2000$, where the effective multipole of each bin is taken as the geometric mean of its edges. The full 3×2 pt data vector comprises all unique tomographic auto- and cross-correlations for cosmic shear, galaxy–galaxy lensing, and galaxy clustering, including all contributions listed in Table 1.

Covariance matrices are computed with the `OneCovariance` code [88], which evaluates Gaussian, non-Gaussian (connected trispectrum), and super-sample covariance contributions using a halo-model framework. The input angular power spectra for the covariance calculation are evaluated at 101 logarithmically spaced multipoles over the same ℓ -range, with 20 logarithmically spaced bins used for each of the lensing and clustering observables. The non-linear power spectrum within `OneCovariance` is computed using `HMCode2020` with the same baryonic-feedback parameter as above. The resulting covariance matrices provide a realistic, scale-dependent error budget that allows the accuracy of `LimberCloud` to be assessed against the expected statistical uncertainties of the survey.

4 Piecewise Linearisation of Physical Quantities

Before arriving at the tensorised expression for the angular power spectra in Section 5, we first express each factor inside the Limber integral (Equation 2.1) in piecewise-linear form on the discrete redshift grid defined in Section 3.3. This section presents the linearisation of the three-dimensional power spectrum (Section 4.1), the number-density kernel (Section 4.2), and the lensing convergence kernel (Section 4.3), together with their validation against CCL. The central result is that both kernel functions can be decomposed as linear combinations of the discretised tomographic redshift distribution values ϕ_i^a , with coefficients that depend only on cosmology and the redshift grid. Under the spatially flat cosmology adopted in Section 3.1, we have $f_K(\chi) = \chi$ throughout; the extension to curved geometries, which amounts to linearising $f_K(\chi)$ itself, is deferred to future work.

4.1 Three-dimensional power spectrum

Within each redshift interval $[\chi_n, \chi_{n+1}]$, the three-dimensional matter power spectrum evaluated at the Limber wavenumber $k = (\ell + 1/2)/\chi$ is represented by a piecewise-linear interpolation between the values at the interval endpoints,

$$P_{uv} \left[\frac{\ell + 1/2}{\chi}, z(\chi) \right] = \frac{\chi_{n+1} - \chi}{\chi_{n+1} - \chi_n} P_{uv}^{\ell,n} + \frac{\chi - \chi_n}{\chi_{n+1} - \chi_n} P_{uv}^{\ell,n+1}, \quad (4.1)$$

where $P_{uv}^{\ell,n} \equiv P_{uv}[(\ell + 1/2)/\chi_n, z(\chi_n)]$. This is the simplest interpolation that preserves continuity across interval boundaries and, crucially, yields integrands that can be evaluated in closed form when combined with the linearised kernel functions derived below.

Figure 2 compares this piecewise-linear representation against direct evaluation by CCL for four representative multipoles, $\ell = 10, 100, 1000$, and $10\,000$. In each panel, the upper sub-panel shows the power spectrum as a function of comoving distance, while the lower sub-panel displays the ratio of each interpolation scheme to the CCL baseline. The `LimberCloud` piecewise-linear scheme agrees with CCL to better than one per cent over the comoving-distance range $\chi \gtrsim 200$ Mpc at all multipoles tested. Deviations exceeding a few per cent are

$\Delta z = 0.01$ and 20 log bins in ℓ . Interpolation is done in χ which connects k and ℓ . what are the k values at which P is evaluated at?

We are normally not used to seeing Power Spectra in this format. Rather $P(k)$ vs k and separate lines for z . How does it look like if you plot it like that?

The differences between the interpolation schemes they are not super clear. Maybe they are not all needed.

confined to very small comoving distances ($\chi \lesssim 200$ Mpc), where the mapping $k = (\ell + 1/2)/\chi$ pushes the wavenumber into the deeply non-linear regime and the power spectrum varies most rapidly with distance. The first-, second-, and third-order numerical interpolation schemes reduce these residuals marginally but would break the analytic integrability on which the tensorised reformulation in Section 5 relies. Since the kernel functions are negligibly small at these extreme distances for all tomographic bins considered, the impact on the final angular power spectra is negligible (see Section 5).

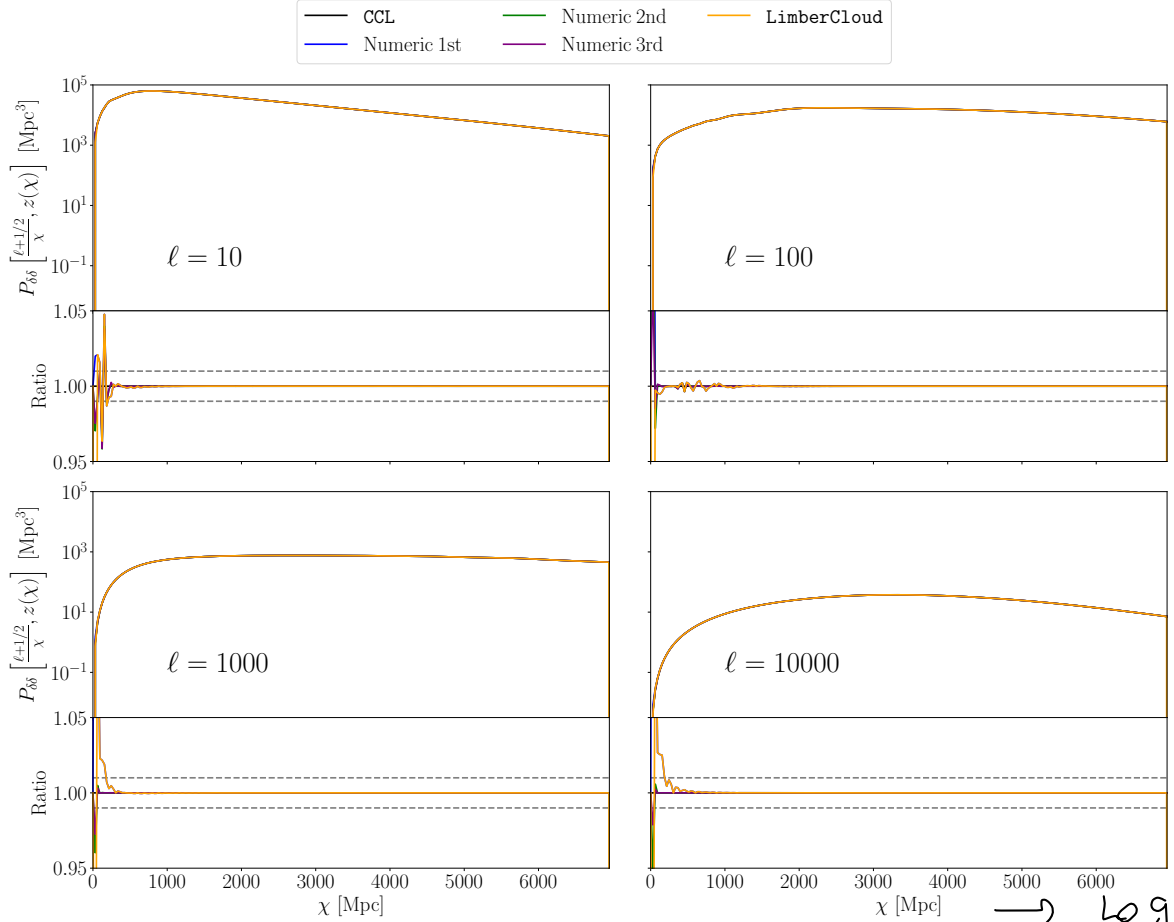


Figure 2. Piecewise-linear interpolation of the non-linear matter power spectrum $P_{\delta\delta}[(\ell + 1/2)/\chi, z(\chi)]$ as a function of comoving distance for four representative multipoles ($\ell = 10, 100, 1000$, and $10\,000$; top-left to bottom-right). In each panel, the upper sub-panel compares the power spectrum evaluated by CCL (solid) with four interpolation schemes identified in the legend: first-order numerical interpolation, second-order numerical interpolation, third-order numerical interpolation, and the LimberCloud piecewise-linear scheme. The lower sub-panel shows the ratio of each scheme to CCL; dashed lines mark the $\pm 1\%$ level.

4.2 Number-density kernel

The number-density kernel $W_{\phi}^a(\chi)$ is determined entirely by the tomographic redshift distribution. Following Equation 2.3, the distribution in comoving-distance space is related to its

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redshift-space counterpart by

$$\phi_n^a \equiv \phi^a(\chi_n) = \frac{H(z_n)}{c} \phi^a(z_n), \quad (4.2)$$

where $n = 0, 1, \dots, N_Z$ indexes the $N_Z + 1$ grid points of the uniform redshift grid defined in Section 3.3, spanning $N_Z = 350$ intervals. Between adjacent grid points, the distribution is reconstructed by linear interpolation,

$$\phi^a(\chi) = \frac{\chi_{n+1} - \chi}{\chi_{n+1} - \chi_n} \phi_n^a + \frac{\chi - \chi_n}{\chi_{n+1} - \chi_n} \phi_{n+1}^a, \quad \chi_n \leq \chi < \chi_{n+1}, \quad (4.3)$$

with the convention that $\phi^a(\chi) = 0$ for $\chi < \chi_0$ and $\chi > \chi_{N_Z}$.

Since the number-density kernel coincides with the redshift distribution itself, it takes the same piecewise-linear form within each interval. We denote by $W_\phi^{a,n}(\chi)$ the restriction of the kernel to the n -th comoving-distance interval $[\chi_n, \chi_{n+1}]$; this superscript convention applies to all interval-restricted quantities throughout this work. The restricted kernel can be written as a weighted sum over all grid-point values,

$$W_\phi^{a,n}(\chi) = \sum_{i=0}^{N_Z} r_\phi^{i,n}(\chi) \phi_i^a, \quad (4.4)$$

where the interpolation coefficients $r_\phi^{i,n}(\chi)$ are non-zero only for $i = n$ and $i = n + 1$:

$$r_\phi^{i,n}(\chi) = \begin{cases} \frac{\chi_{n+1} - \chi}{\chi_{n+1} - \chi_n}, & i = n, \\ \frac{\chi - \chi_n}{\chi_{n+1} - \chi_n}, & i = n + 1, \\ 0, & \text{otherwise.} \end{cases} \quad (4.5)$$

The sparsity of these coefficients—only two non-zero entries per interval—reflects the local support of the piecewise-linear basis and is a key ingredient enabling the efficient tensor contraction in Section 5.

Figure 3 validates the piecewise-linear reconstruction of the number-density kernel against CCL for all five tomographic bins of the LSST Y1 survey scenario. Each panel presents a single bin ($a = 1$ to 5), with the upper sub-panel comparing the kernel evaluated by CCL against four interpolation schemes—first-, second-, and third-order numerical interpolation, together with the LimberCloud piecewise-linear reconstruction—and the lower sub-panel displaying the ratio to CCL. The LimberCloud scheme reproduces the number-density kernels to sub-per-cent accuracy across the full comoving-distance range where the kernel amplitude is non-negligible. The higher-order numerical interpolation schemes exhibit larger spurious oscillations in the ratio panels, confirming that the piecewise-linear approach offers a favourable trade-off between accuracy and analytic tractability.

4.3 Lensing convergence kernel

The lensing convergence kernel $W_\kappa^a(\chi)$ defined in Equation 2.4 involves a line-of-sight integral over the source distribution weighted by the lensing efficiency factor $(\chi' - \chi)/\chi'$. Under the flat-sky assumption, this reads

$$W_\kappa^a(\chi) = \frac{3\Omega_m H_0^2}{2c^2} \frac{\chi}{a(\chi)} \int_\chi^{\chi_H} d\chi' \phi^a(\chi') \frac{\chi' - \chi}{\chi'}. \quad (4.6)$$

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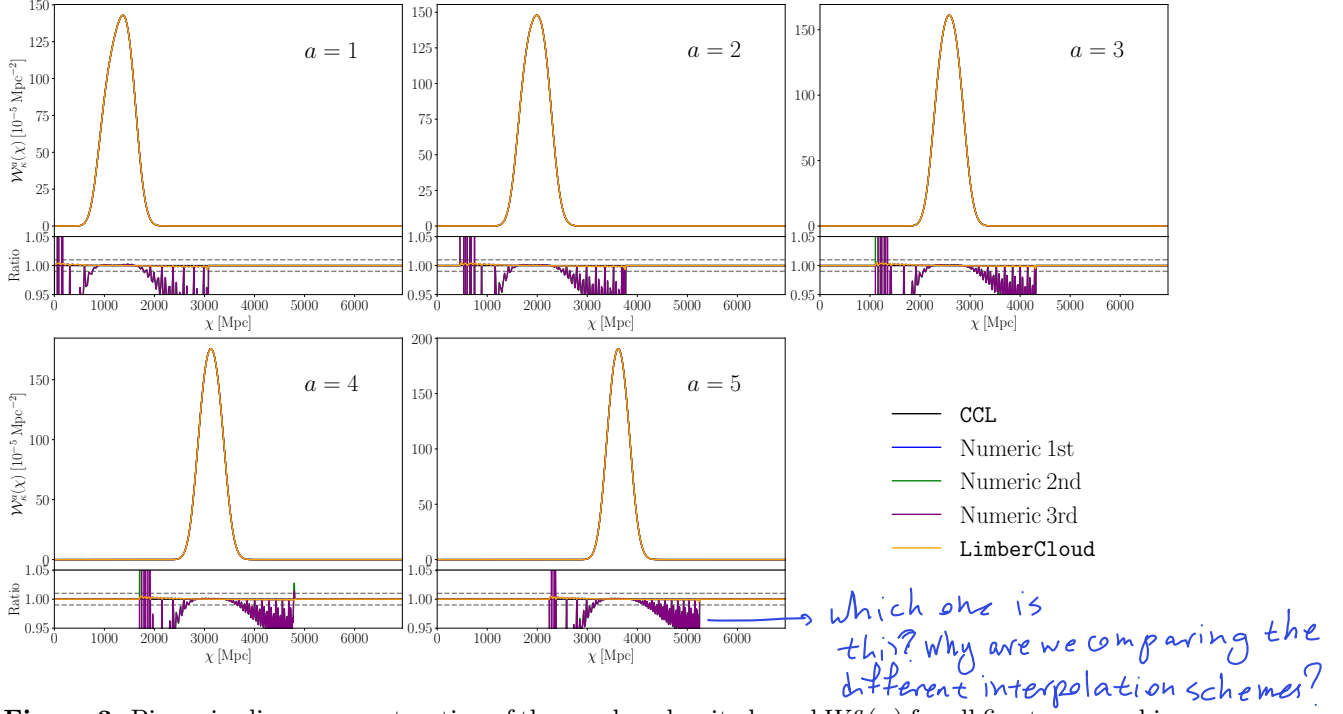


Figure 3. Piecewise-linear reconstruction of the number-density kernel $W_{\phi}^a(\chi)$ for all five tomographic bins ($a = 1$ to 5) of the LSST Y1 survey scenario, ~~arranged in a 2×3 grid with the shared legend in the bottom-right position.~~ In each panel, the upper sub-panel compares the kernel evaluated by CCL (solid) with four interpolation schemes as identified in the legend. The lower sub-panel shows the ratio of each scheme to CCL; dashed lines mark the $\pm 1\%$ level.

When χ lies in the n -th interval $[\chi_n, \chi_{n+1}]$, the integral is split into sub-intervals aligned with the redshift grid,

$$W_{\kappa}^{a,n}(\chi) = \frac{3\Omega_m H_0^2}{2c^2} \frac{\chi}{a(\chi)} \left[\int_{\chi}^{\chi_{n+1}} + \sum_{j=n+1}^{N_Z-1} \int_{\chi_j}^{\chi_{j+1}} \right] d\chi' \phi^a(\chi') \frac{\chi' - \chi}{\chi'}. \quad (4.7)$$

Substituting the piecewise-linear representation of $\phi^a(\chi')$ from Equation 4.3 into each sub-integral and collecting all contributions, the kernel can be expressed as a linear combination of the discretised distribution values,

$$W_{\kappa}^{a,n}(\chi) = \frac{3\Omega_m H_0^2}{2c^2} \frac{\chi}{a(\chi)} \sum_{i=0}^{N_Z} s_{\kappa}^{i,n}(\chi) \phi_i^a. \quad (4.8)$$

Unlike the number-density case, the coefficients $s_{\kappa}^{i,n}(\chi)$ are non-zero for all $i \geq n$, reflecting the non-local nature of the lensing kernel: matter at comoving distance χ receives lensing contributions from all source galaxies at $\chi' > \chi$. Each sub-integral involves a product of two linear functions of χ' divided by χ' , which admits a closed-form evaluation involving

logarithmic terms. The coefficients vanish for $i < n$ and are given for each remaining case by

$$s_{\kappa}^{i=n,n}(\chi) = \frac{\chi_{n+1}^2 - \chi^2}{2(\chi_{n+1} - \chi_n)} - \frac{\chi \chi_{n+1}}{\chi_{n+1} - \chi_n} \ln \frac{\chi_{n+1}}{\chi}, \quad (4.9)$$

$$s_{\kappa}^{i=n+1,n}(\chi) = \frac{\chi_{n+2} - \chi_{n+1}}{2} - \frac{\chi \chi_{n+2}}{\chi_{n+2} - \chi_{n+1}} \ln \frac{\chi_{n+2}}{\chi_{n+1}} + \frac{\chi_{n+1}^2 - 2\chi_n \chi_{n+1} + \chi^2}{2(\chi_{n+1} - \chi_n)} + \frac{\chi \chi_n}{\chi_{n+1} - \chi_n} \ln \frac{\chi_{n+1}}{\chi}, \quad (4.10)$$

$$s_{\kappa}^{n+1 < i < N_Z, n}(\chi) = \frac{\chi_{i+1} - \chi_i}{2} - \frac{\chi \chi_{i+1}}{\chi_{i+1} - \chi_i} \ln \frac{\chi_{i+1}}{\chi_i} + \frac{\chi_i - \chi_{i-1}}{2} + \frac{\chi \chi_{i-1}}{\chi_i - \chi_{i-1}} \ln \frac{\chi_i}{\chi_{i-1}}, \quad (4.11)$$

$$s_{\kappa}^{i=N_Z, n}(\chi) = -\chi + \frac{\chi_{N_Z} - \chi_{N_Z-1}}{2} + \frac{\chi \chi_{N_Z-1}}{\chi_{N_Z} - \chi_{N_Z-1}} \ln \frac{\chi_{N_Z}}{\chi_{N_Z-1}}. \quad (4.12)$$

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The $i = n$ case involves an integral starting from χ (the evaluation point) rather than from a grid boundary, which introduces χ -dependent terms; all other cases integrate over complete grid intervals and depend on χ only through the lensing efficiency factor $(\chi' - \chi)/\chi'$.

The remaining factor $1/a(\chi) = 1 + z(\chi)$ is itself linearised within each interval as

$$1 + z(\chi) = 1 + \frac{\chi_{n+1} - \chi}{\chi_{n+1} - \chi_n} z_n + \frac{\chi - \chi_n}{\chi_{n+1} - \chi_n} z_{n+1}, \quad \chi_n \leq \chi < \chi_{n+1}. \quad (4.13)$$

Absorbing this factor together with the cosmological prefactor into a single set of coefficients, the full lensing convergence kernel takes the same structural form as the number-density kernel,

$$W_{\kappa}^{a,n}(\chi) = \sum_{i=0}^{N_Z} r_{\kappa}^{i,n}(\chi) \phi_i^a, \quad (4.14)$$

where

$$r_{\kappa}^{i,n}(\chi) = \frac{3\Omega_m H_0^2}{2c^2} \chi \left(1 + \frac{\chi_{n+1} - \chi}{\chi_{n+1} - \chi_n} z_n + \frac{\chi - \chi_n}{\chi_{n+1} - \chi_n} z_{n+1} \right) s_{\kappa}^{i,n}(\chi). \quad (4.15)$$

Figure 4 validates the piecewise-linear reconstruction of the lensing convergence kernel against CCL for all five tomographic bins of the LSST Y1 survey scenario. The panel layout and line conventions follow Figure 3. The LimberCloud scheme reproduces the lensing convergence kernels with residuals remaining below one per cent throughout the comoving-distance range that dominates the Limber integral. The oscillatory residuals visible at large comoving distances in the ratio panels—most pronounced for the higher-order numerical schemes—arise where the kernel amplitude has fallen to a negligible fraction of its peak value and do not affect the final angular power spectra. The broader support of W_{κ}^a compared with W_{ϕ}^a reflects the non-local nature of the lensing efficiency integral, which accumulates contributions from all source galaxies beyond the evaluation point.

The central result of this section is that both kernel functions admit the same structural decomposition: within each redshift interval, the kernel for tomographic bin a is a linear function of the discretised distribution values ϕ_i^a , with cosmology- and geometry-dependent coefficients $r_{\phi}^{i,n}$ or $r_{\kappa}^{i,n}$ that are independent of the survey-specific redshift distributions. This clean separation is the foundation of the tensorised reformulation presented in Section 5.

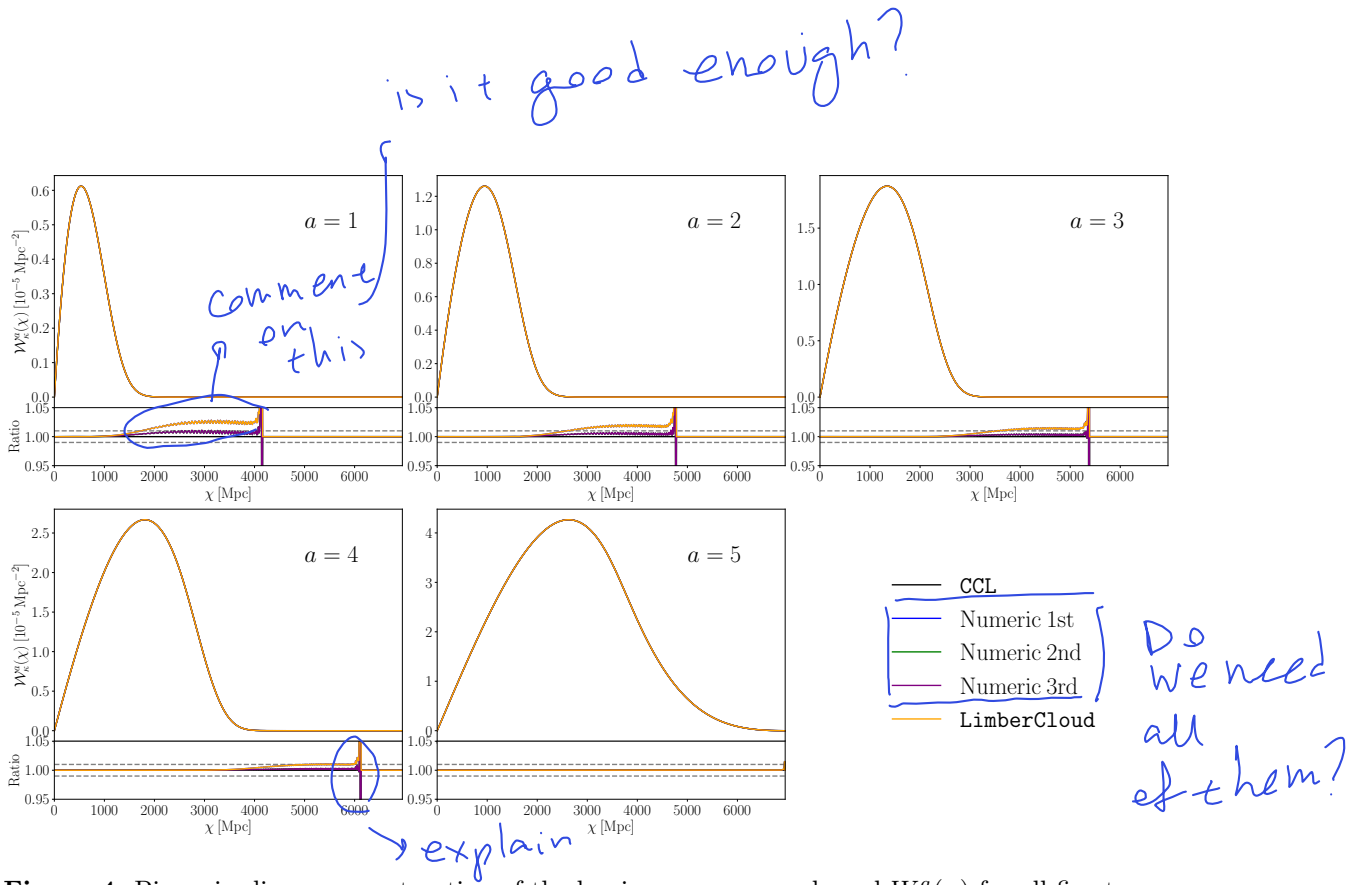


Figure 4. Piecewise-linear reconstruction of the lensing convergence kernel $W_{\kappa}^a(\chi)$ for all five tomographic bins ($a = 1$ to 5) of the LSST Y1 survey scenario, ~~arranged in a 2×3 grid with the shared legend in the bottom-right position.~~ In each panel, the upper sub-panel compares the kernel evaluated by CCL (solid) with four interpolation schemes as identified in the legend. The lower sub-panel shows the ratio of each scheme to CCL; dashed lines mark the $\pm 1\%$ level.

5 Tensorised Reformulation of the Limber Projection

With the piecewise-linear representations of the power spectrum and the kernel functions established in Section 4, we now cast the Limber integral (Equation 2.1) into a tensorised form that cleanly separates cosmology-dependent integrals from survey-dependent redshift distributions. This factorisation is the central result of the paper and the foundation of the LimberCloud framework.

The first step is to decompose the continuous line-of-sight integral into a sum of integrals over the N_Z comoving-distance intervals $[\chi_n, \chi_{n+1}]$ defined by the redshift grid. For a generic tracer pair (u, v) in tomographic bins (a, b) , Equation 2.1 becomes

$$C_{uv}^{ab}(\ell) = A_{uv}(\ell) \sum_{n=0}^{N_Z-1} \int_{\chi_n}^{\chi_{n+1}} d\chi \frac{W_u^{a,n}(\chi) W_v^{b,n}(\chi)}{\chi^2} P_{uv} \left[\frac{\ell + 1/2}{\chi}, z(\chi) \right], \quad (5.1)$$

where the flat-sky simplification $f_K(\chi) = \chi$ has been applied and the superscript n on each kernel restricts it to the n -th comoving-distance interval, following the convention introduced in Section 4.2. The multipole-dependent prefactor $A_{uv}(\ell)$ accounts for the spin of the projected fields and is specified for each tracer pair in Table 1.

The key insight of the tensorised reformulation is that, within each interval, both kernels admit the decomposition derived in Section 4. For the number-density kernel and the lensing

what do you mean by admit?

convergence kernel respectively,

$$W_{\phi}^{a,n}(\chi) = \sum_{i=0}^{N_Z} r_{\phi}^{i,n}(\chi) \phi_i^a, \quad W_{\kappa}^{a,n}(\chi) = \sum_{i=0}^{N_Z} r_{\kappa}^{i,n}(\chi) \phi_i^a, \quad (5.2)$$

where the coefficients $r_{\phi}^{i,n}$ and $r_{\kappa}^{i,n}$ depend only on cosmology and the redshift grid, while the discretised redshift distributions ϕ_i^a encode all survey-specific information. Denoting the kernel type associated with tracer u generically by $r_u^{i,n}$ —standing for either $r_{\phi}^{i,n}$ or $r_{\kappa}^{i,n}$ as specified in Table 1—substituting Equation 5.2 together with the linearised power spectrum (Equation 4.1) into Equation 5.1 yields

$$C_{uv}^{ab}(\ell) = A_{uv}(\ell) \sum_{n=0}^{N_Z-1} \sum_{i=0}^{N_Z} \sum_{j=0}^{N_Z} \left[\int_{\chi_n}^{\chi_{n+1}} d\chi \frac{r_u^{i,n}(\chi) r_v^{j,n}(\chi)}{\chi^2} P_{uv} \left[\frac{\ell + 1/2}{\chi}, z(\chi) \right] \right] \phi_i^a \phi_j^b. \quad (5.3)$$

Since the redshift distribution values ϕ_i^a and ϕ_j^b are independent of both χ and the interval index n , they factor out of the integral and the sum over intervals. This motivates the definition of the *interval tensor*,

$$B_{uv}^{ij,n}(\ell) \equiv A_{uv}(\ell) \int_{\chi_n}^{\chi_{n+1}} d\chi \frac{r_u^{i,n}(\chi) r_v^{j,n}(\chi)}{\chi^2} P_{uv} \left[\frac{\ell + 1/2}{\chi}, z(\chi) \right], \quad (5.4)$$

which encapsulates all cosmology- and geometry-dependent information within a single redshift interval. Summing over intervals defines the *basis tensor*,

$$B_{uv}^{ij}(\ell) \equiv \sum_{n=0}^{N_Z-1} B_{uv}^{ij,n}(\ell), \quad (5.5)$$

so that the angular power spectrum reduces to a compact double contraction,

$$C_{uv}^{ab}(\ell) = \sum_{i=0}^{N_Z} \sum_{j=0}^{N_Z} B_{uv}^{ij}(\ell) \phi_i^a \phi_j^b. \quad (5.6)$$

Equation 5.6 is the principal result of this work. It expresses every angular power spectrum in the 3×2 pt data vector as a bilinear form in the discretised redshift distributions, contracted against a basis tensor $B_{uv}^{ij}(\ell)$ that depends only on cosmological parameters and the redshift grid. Crucially, the basis tensor is independent of the tomographic bin assignments: once $B_{uv}^{ij}(\ell)$ has been computed for a given cosmology and multipole, the angular power spectra for any pair of tomographic bins are obtained by a single matrix–vector contraction with the appropriate redshift distributions. This separation provides a substantial computational advantage whenever the redshift distributions are varied—for instance during self-calibration of photometric redshift uncertainties—because only the outer contraction in Equation 5.6 needs to be re-evaluated, while the expensive integral (Equation 5.4) remains unchanged.

The interval tensor $B_{uv}^{ij,n}(\ell)$ defined in Equation 5.4 involves a product of two piecewise-linear kernel coefficients and the linearised power spectrum, all of which are at most linear in χ within each interval. The resulting integrand is a rational function of χ —with additional logarithmic terms whenever the lensing convergence coefficient $r_{\kappa}^{i,n}$ participates—that admits a

closed-form evaluation in every case. All closed-form expressions were derived using computer algebra (**Mathematica**) and independently verified against direct numerical quadrature.

The structure of $B_{uv}^{ij,n}(\ell)$ is determined by which kernel types appear on each leg of the tracer pair (Table 1), giving rise to three structural cases, each characterised by a distinct set of closed-form integral formulas. When both kernels are of number-density type ($r_\phi \times r_\phi$), the sparsity of $r_\phi^{i,n}$ —non-zero only for $i \in \{n, n+1\}$ (Equation 4.5)—limits the non-vanishing entries to three structurally distinct index configurations per interval, $(i, j) \in \{(n, n), (n, n+1), (n+1, n+1)\}$, after accounting for the symmetry $B_{uv}^{ij,n} = B_{uv}^{ji,n}$. When one kernel is of number-density type and the other of lensing convergence type ($r_\phi \times r_\kappa$), the first index retains the two-point support while the second spans the four structural cases of $r_\kappa^{j,n}$ (Equations 4.9–4.12), yielding $2 \times 4 = 8$ distinct analytic formulas per interval. Because the general structural case (Equation 4.11) covers all intermediate grid points $n+1 < j < N_z$ with a single functional form, these eight formulas suffice to evaluate the $\mathcal{O}(N)$ non-zero entries of the mixed tensor at each interval. When both kernels are of lensing convergence type ($r_\kappa \times r_\kappa$), both indices span the four structural cases, producing ten unique formulas after symmetry reduction. In total, 21 structurally distinct analytic integral formulas per redshift interval are required to populate the full basis tensor for the complete 3×2 pt data vector; each is implemented as a separate function in **LimberCloud**.

The bin-independence of the basis tensor relies on the assumption that the three-dimensional power spectrum, $P_{uv}(k, z)$, and hence any bias functions it absorbs (Table 1), does not depend on the tomographic bin assignment. Three regimes can be distinguished. First, if the bias function is determined entirely by (k, z) and is the same for all bins—as is the case for intrinsic alignments under the non-linear alignment (NLA) model, where $b_I(z) \propto -A_{\text{IA}} C_1 \rho_{\text{cr}} \Omega_m / D(z)$ —the factorisation holds exactly and the basis tensor is computed once per tracer pair. Second, if the bias varies across tomographic bins but is constant within each bin—the standard parameterisation for galaxy bias, which assigns a free amplitude b_g^a to each lens bin—the per-bin constants factor out of the integral and can be absorbed into modified distributions $\tilde{\phi}_i^a = b_g^a \phi_i^a$, so that the basis tensor again needs to be evaluated only once (using $P_{\delta\delta}$ in place of $b_g^2 P_{\delta\delta}$); the magnification response b_μ^a , being a constant per lens bin, is treated identically. Third, if the bias depends on both (k, z) and varies across bins—for example a scale-dependent galaxy bias $b_g^a(k, z)$ that differs between tomographic selections—the bias cannot be factored out of Equation 5.4 and a separate basis tensor must be computed for each bin pair, substantially increasing the computational cost. The first two regimes encompass the bias models most commonly adopted in current Stage-IV 3×2 pt analyses, including those assumed throughout this work (Section 3). Analyses that incorporate beyond-linear, scale-dependent galaxy bias models varying across tomographic selections would enter the third regime; such models are not yet standard but may become relevant as survey precision increases. In either case, the tensorised factorisation remains applicable—only the computational cost of evaluating the basis tensor changes, as discussed further in Section 6.2.

We now validate the tensorised formulation for each of the three combined observables defined in Equations 2.5–2.7: the shape–shape correlation $C_{\epsilon\epsilon}^{ab}(\ell)$, the position–shape correlation $C_{\theta\epsilon}^{ab}(\ell)$, and the position–position correlation $C_{\theta\theta}^{ab}(\ell)$. In each case, the full observable is obtained by computing the basis tensor $B_{uv}^{ij}(\ell)$ for every constituent tracer pair listed in Table 1 and summing the resulting angular power spectra according to Equations 2.5–2.7. The combined spectra are then compared with the direct numerical evaluation by CCL.

* Explain the reason for the other interpolation methods and what they are.

5.1 Shape–shape correlation

The shape–shape angular power spectrum $C_{\ell\ell}^{ab}(\ell)$ receives contributions from gravitational lensing, intrinsic alignments, and their cross-terms (Equation 2.5). For the dominant pure-lensing contribution $C_{\gamma\gamma}^{ab}(\ell)$, both kernels are of lensing convergence type and the spin-2 prefactor is $A_{\gamma\gamma}(\ell) = (\ell + 2)!/[(\ell - 2)!(\ell + 1/2)^4]$. The interval tensor is

$$B_{\gamma\gamma}^{ij,n}(\ell) = A_{\gamma\gamma}(\ell) \int_{\chi_n}^{\chi_{n+1}} d\chi \frac{r_{\kappa}^{i,n}(\chi) r_{\kappa}^{j,n}(\chi)}{\chi^2} P_{\delta\delta} \left[\frac{\ell + 1/2}{\chi}, z(\chi) \right], \quad (5.7)$$

which is dense in both indices, requiring ten unique coefficient elements per interval. The intrinsic-alignment cross-terms $C_{\gamma I}^{ab}(\ell)$ and $C_{I\gamma}^{ab}(\ell)$ each involve one lensing kernel and one number-density kernel, with $P_{I\delta}(k, z) = b_I(k, z) P_{\delta\delta}(k, z)$; by construction, $C_{I\gamma}^{ab}(\ell) = C_{\gamma I}^{ba}(\ell)$, so only one of these two contributions needs to be computed independently. The pure intrinsic-intrinsic term $C_{II}^{ab}(\ell)$ uses $r_{\phi}^{i,n}$ on both legs with $P_{II}(k, z) = b_I^2(k, z) P_{\delta\delta}(k, z)$, recovering the maximally sparse kernel structure with three elements per interval.

Figure 5 presents the absolute fractional error $|\delta_{\ell\ell}^{ab}(\ell)|$ of the tensorised shape–shape power spectra relative to CCL for all 25 source-bin combinations of the LSST Y1 survey scenario. In each panel, the dotted horizontal line marks the one per cent level (10^{-2}), and the solid curve shows the diagonal of the Gaussian covariance normalised to the fiducial spectrum, representing the expected per-bin statistical uncertainty for the survey configuration adopted in Section 3. Across all bin pairs, the LimberCloud error remains below the one per cent reference line over the full multipole range $20 \leq \ell \leq 2000$, with typical residuals at the 10^{-3} level; under the adopted survey configuration this corresponds to one to two orders of magnitude below the per-bin Gaussian covariance.

For the auto-spectra (diagonal panels) and nearby cross-bins at intermediate to high redshifts, the fractional error lies at around the 10^{-3} level. The largest residuals among the auto-spectra occur for the lowest-redshift bin pair ($a = 1, b = 1$), where the lensing signal is intrinsically weaker and the intrinsic-alignment contributions— $C_{\gamma I}$, $C_{I\gamma}$, and C_{II} —carry greater relative weight; partial cancellations between these terms and the dominant gravitational signal reduce the total amplitude and amplify the fractional effect of small absolute errors. A similar trend is visible for widely separated source bins (e.g. $a = 1, b = 5$), where the error curves become more structured for the same reason. Notably, all four interpolation schemes track each other closely in every panel, confirming that the residual error is set primarily by the piecewise-linear power-spectrum interpolation (Equation 4.1) rather than by the kernel treatment; this observation holds across all three observables and will not be repeated below. The fractional error is generally flat or mildly increasing with ℓ , consistent with the expectation that the piecewise-linear approximation becomes less faithful at high multipoles where the Limber wavenumber probes more deeply non-linear scales.

5.2 Position–shape correlation

The position–shape angular power spectrum $C_{\theta\ell}^{ab}(\ell)$ correlates the positions of foreground lens galaxies in bin a with the shapes of background source galaxies in bin b (Equation 2.6). For the dominant gravitational lensing contribution $C_{g\gamma}^{ab}(\ell)$, the number-density kernel appears on the lens leg and the lensing convergence kernel on the source leg, together with the spin-dependent prefactor $A_{g\gamma}(\ell) = \sqrt{(\ell + 2)!/(\ell - 2)!} / (\ell + 1/2)^2$. The interval tensor reads

$$B_{g\gamma}^{ij,n}(\ell) = A_{g\gamma}(\ell) \int_{\chi_n}^{\chi_{n+1}} d\chi \frac{r_{\phi}^{i,n}(\chi) r_{\kappa}^{j,n}(\chi)}{\chi^2} P_{g\delta} \left[\frac{\ell + 1/2}{\chi}, z(\chi) \right], \quad (5.8)$$

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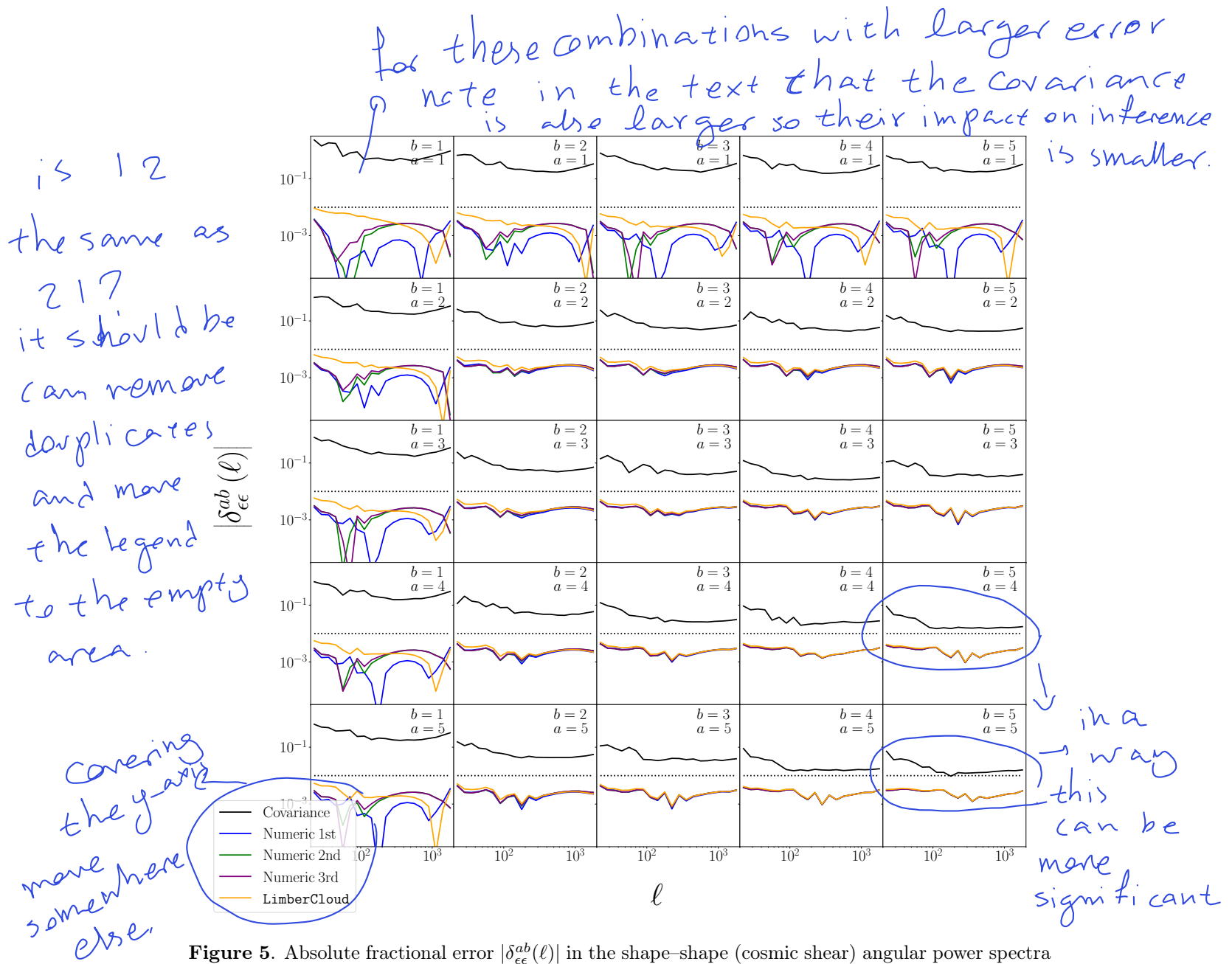


Figure 5. Absolute fractional error $|\delta_{\epsilon\epsilon}^{ab}(\ell)|$ in the shape–shape (cosmic shear) angular power spectra $C_{\epsilon\epsilon}^{ab}(\ell)$ relative to CCL, shown for all 25 source–bin combinations (a, b) of the LSST Y1 survey scenario. Each panel corresponds to a distinct bin pair. The solid curve shows the diagonal of the Gaussian covariance normalised to the fiducial spectrum, representing the expected per-bin statistical uncertainty; the dotted horizontal line marks the one per cent level (10^{-2}). Four interpolation schemes are compared as identified in the legend: first-order numerical interpolation, second-order, third-order, and the LimberCloud piecewise-linear scheme. All schemes remain below the one per cent threshold for all bin combinations, with the largest residuals among auto-spectra occurring at the lowest redshifts where intrinsic-alignment contributions are relatively more important.

where $P_{g\delta}(k, z) = b_g(k, z) P_{\delta\delta}(k, z)$. Since $r_\phi^{i,n}$ is non-zero only for $i \in \{n, n+1\}$ while $r_\kappa^{j,n}$ is non-zero for all $j \geq n$ (Section 4.3), the tensor is sparse in the first index but dense in the second, yielding eight distinct coefficient elements per interval. The intrinsic-alignment contribution $C_{\text{gl}}^{ab}(\ell)$ replaces the lensing kernel with the number-density kernel on the source leg, recovering the maximally sparse three-element structure. The magnification–shear $C_{\mu\gamma}^{ab}(\ell)$ and magnification–intrinsic $C_{\mu\text{I}}^{ab}(\ell)$ contributions follow analogously from the kernel and prefactor assignments in Table 1.

Figure 6 presents the corresponding validation for the position–shape power spectra. For the majority of lens–source bin combinations, the **LimberCloud** error sits at the few $\times 10^{-3}$ level, well below the one per cent reference line; under the adopted survey configuration this corresponds to one to two orders of magnitude below the per-bin Gaussian covariance. A notable exception is the subset of bin pairs in which the lens and source redshift distributions overlap strongly (e.g. $a = 1, b = 1$ and $a = 2, b = 1$), where the fractional error can reach $\sim 10^{-1}$ at low multipoles. This behaviour arises because the gravitational lensing signal $C_{g\gamma}^{ab}$ —which requires source galaxies behind the lenses—is strongly suppressed when the two populations share the same redshift range, leaving the observable dominated by the intrinsic-alignment and magnification contributions whose partial cancellations amplify fractional residuals. In practice, these overlapping bin pairs are not conventionally included in 3×2 pt analyses, which impose a lens–source separation cut on the source leg and (in conjunction with the linear-bias scale cut discussed in Section 5.3) restrict the lens leg to multipoles where the linear-bias model is valid; their statistical uncertainties (solid curves) are correspondingly large. For all standard bin combinations, the **LimberCloud** error is comfortably below both the one per cent line and the survey variance, with a mild uptick at high ℓ visible in certain cross-bins where magnification contributions become relatively more significant.

5.3 Position–position correlation

The position–position angular power spectrum $C_{\theta\theta}^{ab}(\ell)$ receives contributions from intrinsic galaxy–galaxy correlations and from magnification terms (Equation 2.7). For the dominant intrinsic clustering contribution $C_{gg}^{ab}(\ell)$, Table 1 specifies $A_{gg}(\ell) = 1$ and both kernels of number-density type, so that $r_u^{i,n} = r_v^{j,n} = r_\phi^{i,n}$. The interval tensor becomes

$$B_{gg}^{ij,n}(\ell) = \int_{\chi_n}^{\chi_{n+1}} d\chi \frac{r_\phi^{i,n}(\chi) r_\phi^{j,n}(\chi)}{\chi^2} P_{gg} \left[\frac{\ell + 1/2}{\chi}, z(\chi) \right], \quad (5.9)$$

where $P_{gg}(k, z) = b_g^2(k, z) P_{\delta\delta}(k, z)$ absorbs the galaxy bias. Since both kernel coefficients $r_\phi^{i,n}$ are non-zero only for $i \in \{n, n+1\}$ (Equation 4.5), $B_{gg}^{ij,n}$ has at most four non-zero entries per interval—three after symmetry—giving the maximally sparse tensor structure. The magnification cross-terms $C_{g\mu}^{ab}(\ell)$ and $C_{\mu g}^{ab}(\ell)$ each involve one number-density and one lensing convergence kernel, producing an eight-element mixed-sparsity structure, while the magnification auto-term $C_{\mu\mu}^{ab}(\ell)$ uses $r_\kappa^{i,n}$ on both legs and is dense in both indices.

Figure 7 presents the validation for the position–position power spectra. In standard 3×2 pt analyses, only the auto-spectra ($a = b$) are conventionally included, so their accuracy is of primary importance. The auto-spectra are dominated by the intrinsic clustering contribution C_{gg} , which involves only r_ϕ kernels and benefits from the simplest integral structure and maximal sparsity. For the majority of multipoles, the **LimberCloud** error in the auto-spectra lies at around the few times 10^{-3} level. However, at certain multipole ranges the error can approach the diagonal of the survey covariance (solid curve), particularly for the highest-redshift bins where the statistical uncertainties are smallest relative to the signal. This behaviour reflects the sensitivity of galaxy clustering to the precise shape of the redshift distributions: because the number-density kernel W_ϕ^a coincides directly with $\phi^a(\chi)$, any interpolation-induced distortion of the distribution shape propagates without the smoothing provided by the lensing efficiency integral. Nevertheless, this numerical residual remains well below the level of current photometric redshift uncertainties, which dominate the systematic

The main question is are these errors important for parameter estimation and how will they affect it? what level of bias do we expect? One option is to do a quick Fisher forecast for parameter bias. I have a code for that is the KiDS Legacy 3x2pt repo.

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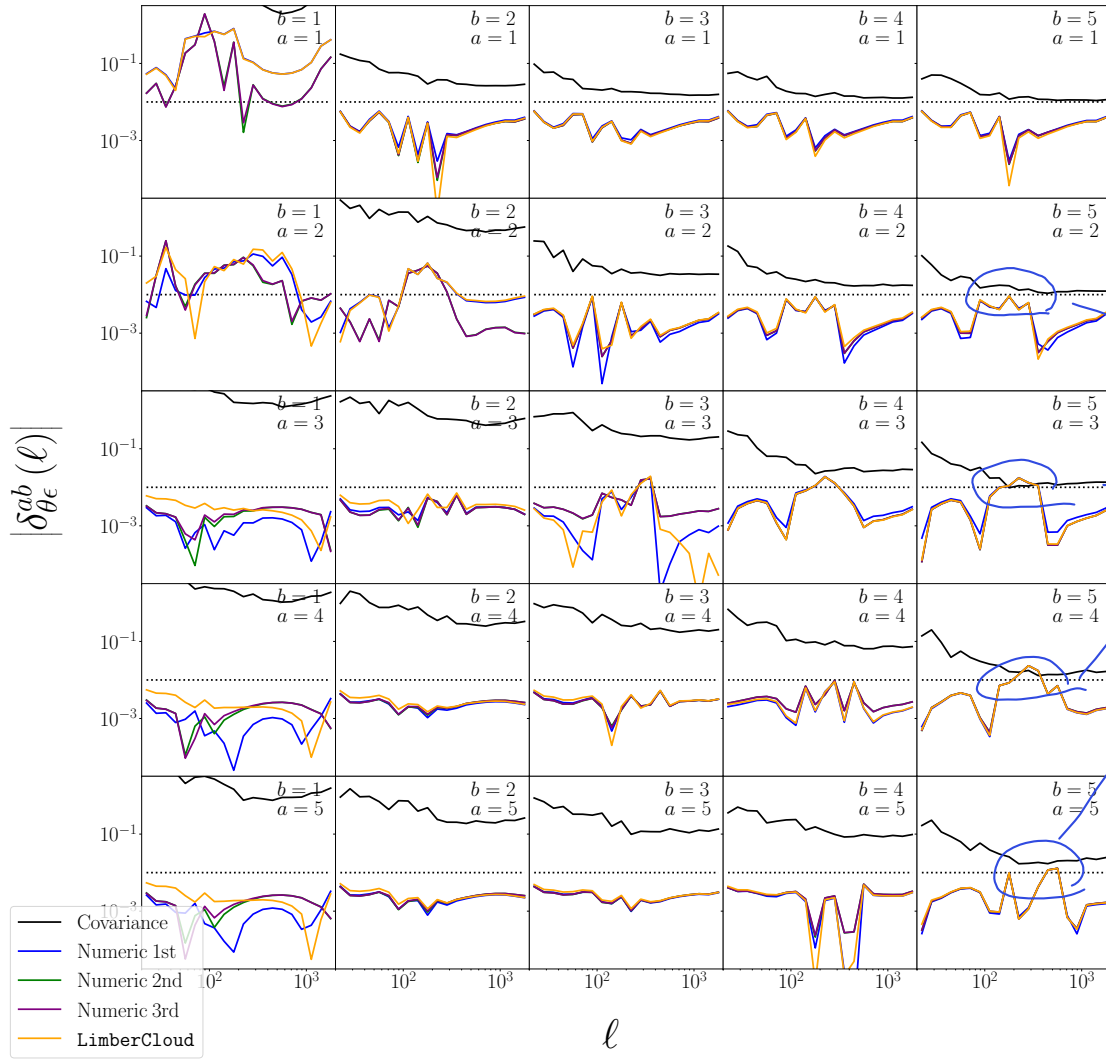


Figure 6. Absolute fractional error $|\delta_{\theta_e}^{ab}(\ell)|$ in the position–shape (galaxy–galaxy lensing) angular power spectra $C_{\theta_e}^{ab}(\ell)$ relative to CCL, shown for all 25 lens–source bin combinations of the LSST Y1 survey scenario. Conventions follow Figure 5. Elevated fractional errors visible in panels where the lens and source bins overlap in redshift (e.g. $a = 1, b = 1$ and $a = 2, b = 1$) arise because the gravitational lensing signal $C_{g\gamma}$ is suppressed when the two populations share the same redshift range, leaving the observable dominated by intrinsic-alignment and magnification contributions; these bin pairs are not conventionally included in 3×2 pt analyses. For all standard bin combinations, the LimberCloud error is comfortably below both the one per cent line and the survey variance.

error budget for galaxy clustering and are themselves marginalised over in cosmological analyses. Furthermore, the multipoles at which the residual is most apparent fall close to or beyond the conventional linear-bias scale cut. The linear galaxy-bias model adopted throughout this work (Section 3.2) is theoretically reliable only for $k \lesssim 0.1 \text{ Mpc}^{-1}$ [41]; at smaller scales, scale-dependent non-linear bias contributions dominate the modelling uncertainty over both the statistical and numerical residuals by a substantial margin. Current Stage-IV 3×2 pt analyses therefore impose a conservative scale cut, conventionally at $k_{\text{max}} \simeq 0.3 h \text{ Mpc}^{-1}$ [19, 24]. Translated per tomographic bin via $\ell_{\text{max}}^a = k_{\text{max}} \chi(z_{\text{up}}^a) - 1/2$ at the upper redshift

which ones do you mean. There hasn't been any data analysis of a stage IV 3x2pt yet.

These can be important for parameter estimation.

at the same time note that the PLK predictions are not 100% accurate. that sets an absolute lower bound on the error.

edge of each lens bin under the fiducial cosmology of Section 3.1, this yields $\ell_{\max} \approx 320$ in the lowest lens bin ($z_{\text{up}} = 0.4$) and $\ell_{\max} \approx 780$ in the highest ($z_{\text{up}} = 1.2$); the residual-covariance crossings visible in Figure 7 lie close to or beyond this excluded regime. The argument is more-over insensitive to the precise cut: even an aggressive alternative such as $k_{\max} \simeq 0.5 \text{ Mpc}^{-1}$, accessible with hybrid effective field theory bias models [42], places the residuals well inside the bias-modelling-dominated regime. The piecewise-linear approximation error is therefore negligible in the context of realistic inference.

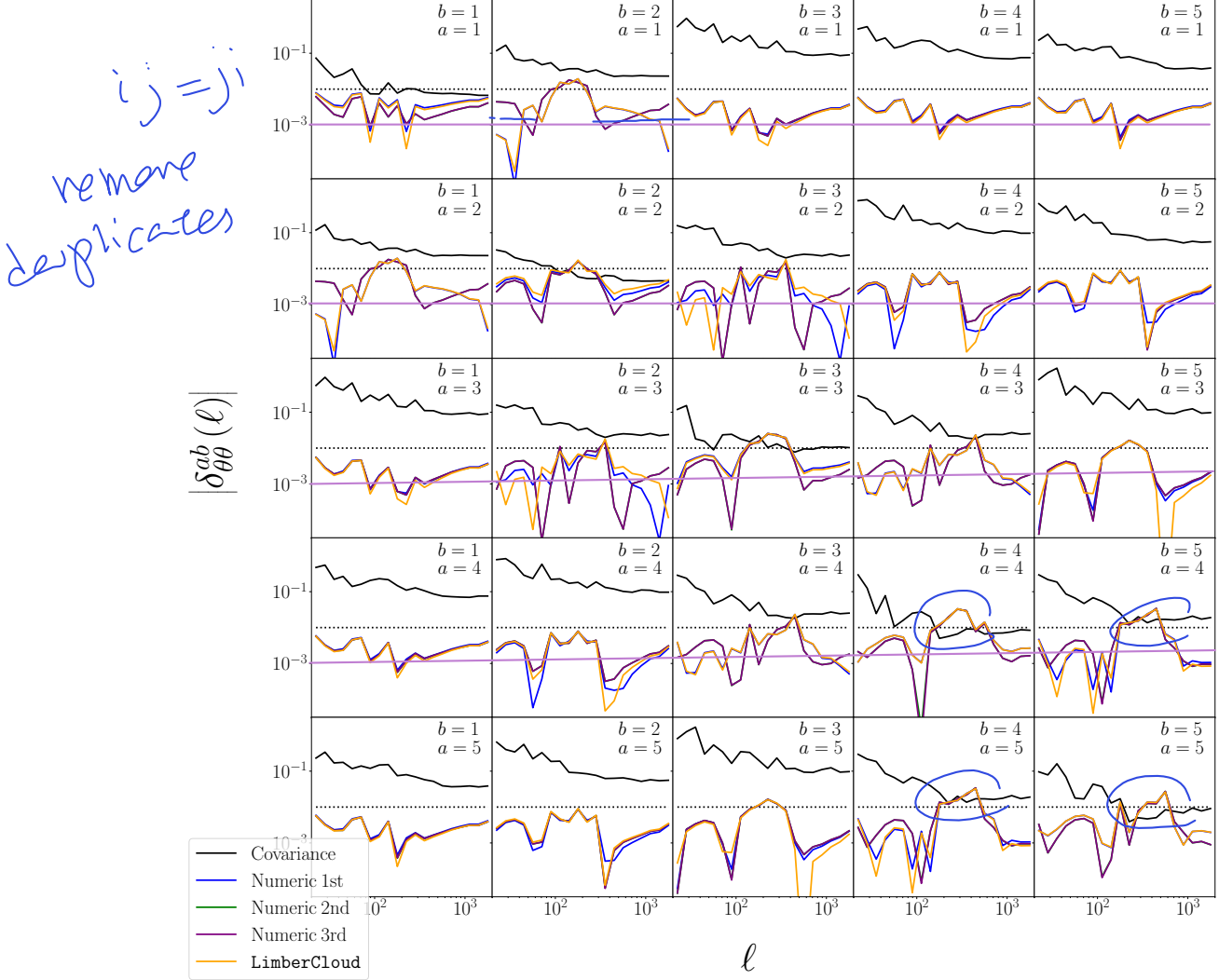


Figure 7. Absolute fractional error $|\delta_{\theta\theta}^{ab}(\ell)|$ in the position-position (galaxy clustering) angular power spectra $C_{\theta\theta}^{ab}(\ell)$ relative to CCL, for all 25 lens-bin combinations of the LSST Y1 survey scenario. Conventions follow Figure 5. The auto-spectra (diagonal panels), dominated by intrinsic galaxy clustering C'_{gg} , typically achieve few times 10^{-3} accuracy; at certain multipole ranges the error can approach the survey covariance level, reflecting the sensitivity of galaxy clustering to the precise shape of the redshift distributions. Cross-bin spectra for widely separated lens bins (e.g. $a = 1, b = 5$) exhibit larger fractional errors because the observable is dominated by subdominant magnification contributions ($C'_{\text{g}\mu}, C'_{\mu\text{g}}, C'_{\mu\mu}$); these spectra are not conventionally included in standard cosmological analyses and their statistical uncertainties (solid curve) are correspondingly large.

would finer Δz binning fix this problem?



The off-diagonal panels ($a \neq b$) exhibit larger fractional errors for widely separated lens bins (e.g. $a = 1, b = 5$), reaching $\sim 10^{-2}$ – 10^{-1} at certain multipoles. This occurs because the intrinsic clustering signal C_{gg} is negligible for non-overlapping bins, leaving the observable dominated by the much fainter magnification contributions ($C_{g\mu}, C_{\mu g}, C_{\mu\mu}$). These cross-bin spectra are not conventionally included in cosmological analyses and their statistical uncertainties are correspondingly large.

6 Evaluation and Discussion

The preceding section demonstrated that the tensorised formulation reproduces the angular power spectra computed by CCL to sub-per-cent accuracy across all observables and bin combinations relevant for the LSST Y1 survey scenario. Having established this numerical fidelity, we now turn to the practical implications of the framework for Stage-IV cosmological inference. This section addresses four complementary aspects: the computational performance of the tensorised implementation relative to conventional Limber codes (Section 6.1), the prospect of end-to-end emulation of the cosmology-dependent coefficients (Section 6.2), the capacity for analytic marginalisation over redshift distribution uncertainties (Section 6.3), and natural extensions to non-flat geometries and beyond-Limber scales (Section 6.4).

6.1 Computational performance

A central motivation for the tensorised reformulation is the clean separation of cosmology-dependent computation from the survey-specific projection. To quantify the resulting performance characteristics, we benchmark four implementations of the full 3×2 pt angular power spectrum computation: (i) CCL, which evaluates the Limber integral by numerical quadrature; (ii) Numba-CPU, a LimberCloud implementation accelerated with Numba just-in-time (JIT) compilation on CPU; (iii) JAX-CPU, the LimberCloud implementation compiled and executed on a multi-core CPU via JAX; and (iv) JAX-GPU, the same JAX implementation executed on a single NVIDIA GPU. All four implementations adopt identical cosmological parameters, multipole grids, tomographic binning, and redshift distributions, ensuring a direct comparison.

For the three LimberCloud implementations, the computation decomposes naturally into three clearly separated stages. The *cosmology stage* computes the comoving distance–redshift relation $\chi(z)$ and the non-linear three-dimensional power spectrum $P_{\delta\delta}(k, z)$ using CAMB via the CCL back-end. The *coefficient stage* evaluates the interval tensor $B_{uv}^{ij,n}(\ell)$ (Equation 5.4) for all tracer pairs, grid-point indices, and multipoles using the 21 analytic integral formulas, and sums the result into the basis tensor $B_{uv}^{ij}(\ell)$ (Equation 5.5). The *projection stage* contracts the basis tensor with the discretised redshift distributions via Equation 5.6 to obtain the angular power spectra for all tomographic bin pairs. This three-stage decomposition is a direct consequence of the tensorised structure: the cosmology and coefficient stages depend only on the cosmological parameters and the redshift grid, while the projection stage depends only on the tomographic bin assignments ϕ_i^a and ϕ_j^b . For CCL, no such separation exists; the line-of-sight integration and summation over tomographic bins are performed together within a single monolithic call.

The benchmark consists of a series of sequential evaluations, each with an independently drawn cosmology sampled uniformly within $\pm 10\%$ of the fiducial parameter values listed in Section 3.1. Cumulative wall-clock times are recorded for each stage as the number of evaluations is varied from 100 to 1000 in ten linearly spaced steps. For the JAX implementations,

all timings include the overhead of JIT compilation and, for the GPU configuration, host-to-device memory transfers; explicit device synchronisation is enforced after every computation to ensure that all GPU operations have completed before the timer is read.

Figure 8 presents the benchmark results for the LSST Y1 survey configuration. The left column shows the shape–shape (cosmic shear) observable alone, while the right column shows the full 3×2 pt computation including all three tracer-pair combinations. Within each column, the four panels display, from top to bottom, the total cumulative time and its decomposition into the cosmology, coefficient, and projection stages. In the lower three panels, the CCL curve shows the total CCL time as a fixed reference, since the CCL pipeline does not decompose into these stages; this convention allows the cost of each LimberCloud stage to be compared directly against the full conventional computation. Both configurations exhibit qualitatively consistent scaling behaviour: the cosmology stage dominates the total budget, the coefficient and projection stages are substantially cheaper, and the GPU implementation delivers the largest gains in the projection step. The 3×2 pt case shows proportionally larger absolute times in the coefficient and projection stages, reflecting the increased number of tracer-pair and tomographic-bin combinations, but the relative performance ordering among implementations is preserved.

The cosmology stage dominates the total computation time for all implementations, accounting for more than half of the wall-clock budget at 1000 evaluations. This stage is shared across all four implementations, as it relies on the same CAMB and CCL back-end; the tight clustering of the four curves in the cosmology panel confirms that all differences in total runtime arise from the non-cosmological stages. This observation has an important practical implication: any further acceleration of the total pipeline requires either replacing the cosmology computation with a fast surrogate—as discussed in Section 6.2—or adopting a fast–slow sampling strategy [80–82] that avoids re-evaluating the cosmology stage when only nuisance parameters change.

The coefficient stage reveals the first substantial performance differences among implementations. The JAX–GPU implementation evaluates the 21 analytic integral formulas approximately five times faster than the Numba–CPU alternative at 1000 evaluations. This gain arises because the coefficient evaluation involves independent integrals over $N_z = 350$ redshift intervals for each multipole and tracer pair, a workload that maps naturally onto the massively parallel execution model of GPU hardware.

The projection stage exhibits the most dramatic performance gains. For the full 3×2 pt configuration (Figure 8b), the JAX–GPU implementation completes the projection—contracting the basis tensor with ϕ_i^a and ϕ_j^b for all bin pairs, tracer pairs, and multipoles—in approximately 3 s at 1000 evaluations, compared with approximately 45 s for JAX–CPU and approximately 700 s for Numba–CPU. The shape–shape configuration (Figure 8a) exhibits the same performance hierarchy at proportionally lower absolute cost, consistent with the reduced number of bin-pair combinations. In both cases, the projection step reduces to a batch of bilinear contractions (Equation 5.6), an operation for which GPU linear-algebra libraries are heavily optimised. The JAX–GPU projection is faster than the full CCL pipeline by more than three orders of magnitude and faster than JAX–CPU by approximately one order of magnitude.

In terms of total runtime at 1000 evaluations, the JAX–GPU implementation is approximately twice as fast as CCL, with the gain constrained by the shared cosmology bottleneck. However, this headline figure significantly understates the advantage of the tensorised structure in the setting most relevant for practical inference. In a typical Markov chain Monte Carlo (MCMC) analysis, the parameter space comprises both model parameters $\boldsymbol{\vartheta} = (\boldsymbol{\Omega}, \boldsymbol{\lambda})$ —

This is a bit misleading as the plot is not showing how long it takes for CCL to do the Limber integral.

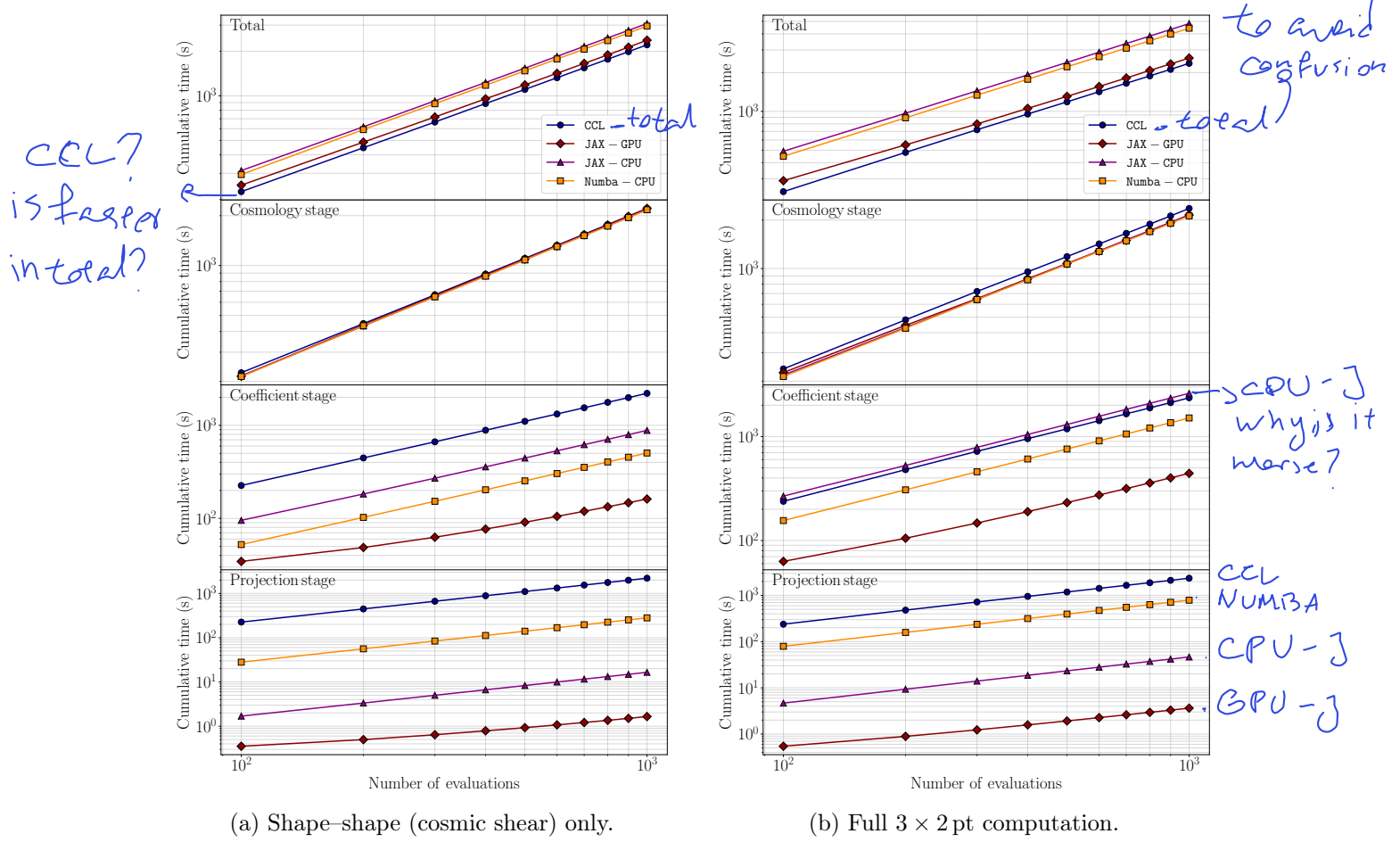


Figure 8. Cumulative wall-clock time as a function of the number of sequential evaluations for four implementations of the angular power spectrum computation in the LSST Y1 survey configuration. *Left:* shape-shape (cosmic shear) observable only. *Right:* full 3×2 pt computation including shape-shape, position-shape, and position-position tracer pairs. Within each column, the panels show, from top to bottom, the total time and its decomposition into the cosmology, coefficient, and projection stages. In each of the lower three panels, the CCL curve (circles) reproduces the total CCL time as a reference, since the conventional pipeline does not separate into these stages. All implementations use identical cosmological parameters, multipole grids, and redshift distributions; each evaluation draws an independent random cosmology within $\pm 10\%$ of the fiducial values.

encompassing cosmological and astrophysical parameters—and nuisance parameters Ξ controlling the redshift distributions (Section 6.3). Modern samplers exploit the observation that nuisance-parameter proposals do not require re-evaluation of the cosmological computation: under the tensorised formulation, proposals that modify only the redshift distributions require only the inexpensive projection step (Equation 5.6), while the basis tensor $B_{uv}^{ij}(\ell)$ is reused from the previous cosmological evaluation. In this fast-proposal regime, the effective cost per likelihood evaluation on JAX-GPU is reduced to the projection time alone—approximately 3 ms per evaluation—representing a speed-up of more than three orders of magnitude over the full CCL pipeline. Even on CPU, the JAX-CPU projection achieves a speed-up of approximately two orders of magnitude. This separation of time-scales provides a natural and efficient interface for fast-slow sampling strategies, substantially reducing the overall cost of high-dimensional

A little wordy here with very long sentences.
Too many conditions and adjectives. Think of simplifying
inference. the language. A few examples given below.

6.2 Prospects for end-to-end emulation

The benchmark results above establish that the cosmology stage—the evaluation of the comoving distance–redshift relation and the non-linear three-dimensional power spectrum via CAMB and HMCODE2020—dominates the total wall-clock time for all implementations. Emulators for the three-dimensional matter power spectrum are already mature: tools such as CosmoPower [89], BACCO [90], EuclidEmulator2 [58], and the Mira-Titan emulator [59] predict $P_{\delta\delta}(k, z)$ to sub-per-cent accuracy in milliseconds, effectively eliminating the Boltzmann-solver bottleneck. A complementary factorisation strategy is the COBRA framework of Bakx et al. [91], which decomposes the three-dimensional matter power spectrum itself into a small set of cosmology-independent basis functions whose amplitudes depend only on cosmological parameters; this approach operates at the level of $P_{\delta\delta}(k, z)$ and is naturally combinable with the present framework, which factorises the projected angular power spectrum one step further along the survey axis and could in principle adopt a COBRA-style surrogate in place of the CAMB back-end in the cosmology stage. The tensorised structure of LimberCloud extends this acceleration one step further: rather than emulating only the three-dimensional power spectrum, it enables *end-to-end emulation* of the basis tensor $B_{uv}^{ij}(\ell)$, bridging the gap between three-dimensional emulation and the projected summary statistics that enter the likelihood.

The basis tensor depends on the full set of model parameters $\boldsymbol{\vartheta} = (\boldsymbol{\Omega}, \boldsymbol{\lambda})$, comprising both cosmological parameters $\boldsymbol{\Omega}$ (e.g. $\Omega_m, \sigma_8, w_0, w_a$) and astrophysical parameters $\boldsymbol{\lambda}$ (e.g. intrinsic-alignment amplitude, galaxy bias values, baryonic feedback strength) that enter through the three-dimensional power spectra $P_{uv}(k, z)$. The emulation strategy proceeds in two phases of offline precomputation. In the first phase, an existing surrogate provides $P_{\delta\delta}(k, z; \boldsymbol{\Omega})$ at negligible cost, eliminating the Boltzmann-solver and halo-model evaluations that dominate the cosmology stage. In the second phase, the emulated power spectrum is combined with the astrophysical bias models to form $P_{uv}(k, z; \boldsymbol{\vartheta})$, and the coefficient stage evaluates $B_{uv}^{ij}(\ell; \boldsymbol{\vartheta})$ via the 21 analytic integral formulas for a large training ensemble of parameter vectors $\boldsymbol{\vartheta}$. A lightweight machine-learning surrogate is then trained to predict $B_{uv}^{ij}(\ell)$ directly for any new $\boldsymbol{\vartheta}$. At inference time, the emulator forward pass replaces both the cosmology and coefficient stages, and the full angular power spectrum for arbitrary tomographic bin pairs is reconstructed via the bilinear contraction of Equation 5.6.

The feasibility of this precomputation can be assessed from the benchmark results of Section 6.1. With the first-phase surrogate in place, only the coefficient stage needs to be evaluated directly, at a cost of approximately 0.5 s per parameter vector on a single GPU for the full 3×2 pt configuration. A training set of 10^6 parameter vectors—of the same order as the number of independent basis-tensor elements—therefore requires approximately six days on a single GPU node. The workload is embarrassingly parallel across independent parameter vectors: distributing over $\mathcal{O}(10)$ single-GPU nodes reduces the wall-clock time to approximately 14 hours.

In the most demanding scenario identified in Section 5, where scale- and bin-dependent biases require a separate basis tensor for each tomographic bin pair, the coefficient cost increases by up to a factor of ~ 25 for the affected tracer pairs. (For 10^6 training vectors this amounts to approximately 100 days on a single GPU, but the cost is readily reduced by combining parallelisation across $\mathcal{O}(10)$ nodes, utilisation of multiple GPUs per node (up to four on typical HPC hardware), and explicit multi-process parallelism within each node in addition to the internal multi-threaded execution already employed by JAX.) Together, these

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strategies can bring the wall-clock time to the order of one day. While the simplest and most widely used bias parameterisations in current Stage-IV analyses fall within the first two regimes of Section 5—for which the basis tensor is computed once per tracer pair—analyses incorporating beyond-linear, scale-dependent galaxy bias models that vary across tomographic selections would enter this third regime. The estimates above confirm that even this worst-case scenario remains accessible to the emulation framework.

The primary advantage of end-to-end emulation is a dramatic reduction in the cost per likelihood evaluation during inference. (The cosmology and coefficient stages, which together account for the vast majority of the wall-clock time (Figure 8), are replaced by a neural-network forward pass whose cost is independent of the complexity of the underlying physical model.) The remaining projection step executes in approximately 3ms per evaluation on GPU hardware (Section 6.1). The combined emulation-plus-projection pipeline is therefore expected to achieve a speed-up of three to four orders of magnitude over the conventional CCL approach, enabling rapid exploration of high-dimensional parameter spaces in both likelihood-based and simulation-based inference.

A further advantage is scalability. The projection step and the emulator forward pass are both embarrassingly parallel across parameter samples and tomographic bin pairs, so the throughput scales linearly with available compute units—additional GPUs translate directly into proportionally more evaluations per unit time. By contrast, the conventional CCL pipeline performs the Limber integration monolithically for each bin pair at each parameter-space point, offering limited scope for hardware-level parallelism beyond internal thread-level concurrency.

The main challenge for end-to-end emulation lies not in the dimensionality of the input parameter space—which is typically 10–50 and is well sampled by 10^6 training vectors—but in the dimensionality of the emulation target: the basis tensor comprises $\sim 10^6$ independent elements, each of which must be predicted to sufficient accuracy across all multipoles and tracer pairs simultaneously. The tensorised representation offers several structural advantages that mitigate this challenge. A large fraction of basis-tensor elements are exactly zero due to the sparsity of the kernel coefficients: for the $r_\phi \times r_\phi$ case, the basis tensor is tridiagonal in (i, j) and more than 99% of its entries vanish; for the mixed $r_\phi \times r_\kappa$ case, the lower-triangular half is zero; and for the $r_\kappa \times r_\kappa$ case, the matrix is dense but symmetric, so only the upper triangle contains independent information. These sparsity patterns and symmetries can be exploited through dimensionality-reduction techniques—such as principal component analysis (PCA) applied to the non-zero entries across the training set—to compress the emulation target to a low-dimensional latent space before training the surrogate model. Additional strategies include hierarchical or probe-specific emulator architectures that exploit the block structure across tracer pairs, and physics-informed network designs that encode known scaling relations among the basis-tensor components. A full demonstration of end-to-end emulation within a differentiable inference pipeline is left to follow-up work.

6.3 Analytic marginalisation over redshift distribution uncertainties

Complementary to the emulation strategy of the preceding subsection—which accelerates the cosmology-dependent computation—the tensorised formulation also enables a fundamentally different treatment of the nuisance parameters controlling the survey-specific redshift distributions. In conventional analyses, uncertainties in the tomographic redshift distributions are explored by sampling: at each MCMC step, the distributions are perturbed and the Limber integrals are re-evaluated in full. The bilinear structure of Equation 5.6 makes it possible to

no need to repeat this.

Some of this is also repeated

There is repetition here. mention once, that PCK emulators exist and are very fast, which makes the projection the bottleneck then focus on that.

This whole section can be substantially decreased. Think of the core message. - 26 - Write it in bullet points first. Then turn it into succinct text.

marginalise over these uncertainties *analytically*, eliminating the need to sample over them and reducing the effective dimensionality of the parameter space.

To formulate the marginalisation, we recast Equation 5.6 in compact matrix notation. For a given tracer pair (u, v) and multipole ℓ , define the basis-tensor matrix $[\mathbf{B}_{uv}(\ell)]_{ij} \equiv B_{uv}^{ij}(\ell)$ and the distribution matrix $\Phi \equiv [\phi^1, \phi^2, \dots, \phi^{N_B}]$, an $(N_Z+1) \times N_B$ matrix whose a -th column is the discretised redshift distribution for tomographic bin a , with N_B denoting the number of tomographic bins. The full set of angular power spectra for all bin pairs is then expressed as the matrix bilinear form

$$\mathbf{C}_{uv}(\ell) = \Phi^T \mathbf{B}_{uv}(\ell) \Phi, \quad (6.1)$$

where $[\mathbf{C}_{uv}(\ell)]_{ab} = C_{uv}^{ab}(\ell)$ is the full matrix of angular power spectra across tomographic bins. This expression is exact and equivalent to Equation 5.6; its bilinear structure in the distribution matrix is the key property that enables the analytic treatment developed below.

We model uncertainties in the tomographic redshift distributions as perturbations about a fiducial model. Writing $\Phi = \bar{\Phi} + \Xi$, where $\bar{\Phi}$ is the fiducial distribution matrix and $\Xi \equiv [\xi^1, \xi^2, \dots, \xi^{N_B}]$ is the perturbation matrix whose a -th column ξ^a represents the departure of the true distribution from the fiducial in bin a . The nuisance parameters over which we seek to marginalise are precisely the perturbation components ξ_i^a . Substituting into Equation 6.1 and expanding algebraically yields

$$\mathbf{C}_{uv}(\ell; \Xi) = \bar{\mathbf{C}}_{uv}(\ell) + \Xi^T \mathbf{B}_{uv}(\ell) \bar{\Phi} + \bar{\Phi}^T \mathbf{B}_{uv}(\ell) \Xi + \Xi^T \mathbf{B}_{uv}(\ell) \Xi, \quad (6.2)$$

where $\bar{\mathbf{C}}_{uv}(\ell) \equiv \bar{\Phi}^T \mathbf{B}_{uv}(\ell) \bar{\Phi}$ is the fiducial angular power spectrum matrix. This expansion is *exact*: because the angular power spectrum is bilinear in the distribution vectors, the polynomial terminates at second order in the perturbations and involves no approximation. This stands in contrast to existing analytic marginalisation approaches based on Taylor expansions of the Limber integral [49, 52–54, 79, 85], which truncate the dependence on the nuisance parameters at first order and thereby introduce a systematic truncation error.

Collecting all angular power spectra across tracer pairs, bin pairs, and multipoles into a theory data vector $\mathcal{M}$, the expansion (Equation 6.2) yields the data-vector decomposition

$$\mathcal{M}(\vartheta, \Xi) = \bar{\mathcal{M}}(\vartheta) + \mathbf{\Gamma}(\vartheta) \Xi + \Xi^T \mathbf{B}(\vartheta) \Xi, \quad (6.3)$$

where ϑ denotes the model parameters (cosmological and astrophysical), $\bar{\mathcal{M}}(\vartheta)$ is the fiducial data vector, $\mathbf{\Gamma}(\vartheta)$ is the linear response matrix encoding the first-order dependence on Ξ , and $\Xi^T \mathbf{B}(\vartheta) \Xi$ denotes the vector whose entries are the bilinear contributions $(\xi^a)^T \mathbf{B}_{uv}(\ell; \vartheta) \xi^b$ for each data-vector element. Both terms follow directly from the algebraic structure of the bilinear form and require no finite-difference approximations. In practice, the high-dimensional nuisance vector Ξ may be projected onto a low-dimensional basis describing physically motivated modes such as coherent redshift shifts and broadening; the formalism applies without modification to any such linear reparameterisation.

We now consider the analytic marginalisation of Ξ under a Gaussian likelihood for the observed data vector $\mathcal{D}$,

$$-2 \ln \mathcal{L}(\mathcal{D} | \vartheta, \Xi) = [\mathcal{D} - \mathcal{M}(\vartheta, \Xi)]^T \Sigma^{-1} [\mathcal{D} - \mathcal{M}(\vartheta, \Xi)], \quad (6.4)$$

where Σ is the data covariance matrix. Defining the fiducial residual $\Delta(\vartheta) \equiv \mathcal{D} - \bar{\mathcal{M}}(\vartheta)$ and substituting Equation 6.3, the quadratic form in the exponent separates into a Gaussian

can you define these more explicitly?

part and higher-order corrections,

$$\begin{aligned}
[\mathcal{D} - \mathcal{M}]^T \Sigma^{-1} [\mathcal{D} - \mathcal{M}] = & \underbrace{\Delta^T \Sigma^{-1} \Delta}_{\mathcal{O}(\Xi^0)} - \underbrace{2 \Delta^T \Sigma^{-1} \Gamma \Xi}_{\mathcal{O}(\Xi^1)} + \underbrace{\Xi^T \Gamma^T \Sigma^{-1} \Gamma \Xi}_{\mathcal{O}(\Xi^2)} \\
& - \underbrace{2 \Delta^T \Sigma^{-1} (\Xi^T \mathcal{B} \Xi)}_{\mathcal{O}(\Xi^2)} + \underbrace{2 \Xi^T \Gamma^T \Sigma^{-1} (\Xi^T \mathcal{B} \Xi)}_{\mathcal{O}(\Xi^3)} \\
& + \underbrace{(\Xi^T \mathcal{B} \Xi)^T \Sigma^{-1} (\Xi^T \mathcal{B} \Xi)}_{\mathcal{O}(\Xi^4)}.
\end{aligned} \tag{6.5}$$

Jose says that the order dependence of each term is written under it.

The first line collects terms of zeroth through second order in Ξ and constitutes a Gaussian quadratic form in the nuisance vector; the subsequent lines arise from cross-terms and self-interactions involving the bilinear contribution $\Xi^T \mathcal{B} \Xi$, spanning second through fourth order.

At linear order in the data vector perturbation—neglecting the bilinear term $\Xi^T \mathcal{B} \Xi$ —the integrand of the marginalisation integral is exactly Gaussian in Ξ . Assuming a Gaussian prior $p(\Xi) = \mathcal{N}(\mathbf{0}, \mathbf{\Pi})$, the marginalised likelihood

$$\mathcal{L}(\mathcal{D} | \vartheta) = \int \mathcal{L}(\mathcal{D} | \vartheta, \Xi) p(\Xi) d\Xi \tag{6.6}$$

is that the second term in 6.5?

is evaluated by completing the square in Ξ and performing the Gaussian integral, yielding

$$-2 \ln \mathcal{L}(\mathcal{D} | \vartheta) = \Delta^T \tilde{\Sigma}^{-1}(\vartheta) \Delta + \ln \det \tilde{\Sigma}(\vartheta), \tag{6.7}$$

where the effective covariance is

Start with this and say that we can recover the first order approach by ignoring all terms above $\mathcal{O}(\Xi^1)$

$$\tilde{\Sigma}(\vartheta) = \Sigma + \Gamma(\vartheta) \mathbf{\Pi} \Gamma(\vartheta)^T. \tag{6.8}$$

Equations 6.7 and 6.8 constitute the *linear-order* analytic marginalisation result: the marginalisation over redshift distribution uncertainties reduces to an inflation of the data covariance by $\mathbf{\Gamma} \mathbf{\Pi} \mathbf{\Gamma}^T$, where $\mathbf{\Gamma}$ is obtained directly from the basis tensor and the fiducial distributions. This result is structurally identical to that of existing first-order approaches [49, 52–54, 79, 85]; the key distinction is that $\mathbf{\Gamma}$ is read off from the exact bilinear expansion (Equation 6.2) rather than estimated by finite differencing of the Limber integral.

We note that the bilinear contribution $\Xi^T \mathcal{B} \Xi$, neglected in the linear-order result above, introduces non-Gaussian terms in the marginalisation integral through the second line of Equation 6.5. While these higher-order corrections can in principle be evaluated systematically—for instance using Isserlis’ theorem [92] to compute the required polynomial moments of the Gaussian prior—they are expected to be small when the prior covariance $\mathbf{\Pi}$ is well constrained by photometric redshift calibration, as the corrections are suppressed by additional powers of $\|\xi^a\|/\|\bar{\phi}^a\|$ relative to the linear-order result. The tensorised formulation nevertheless provides these higher-order terms from first principles, offering a controlled framework in which the adequacy of the linear-order approximation can be assessed quantitatively. A detailed investigation of these contributions, together with numerical validation within a full inference pipeline, is left to follow-up work.

This is important more up.

Compared to existing analytic marginalisation approaches, the tensorised formulation offers three key advantages. First, the dependence of the data vector on the nuisance parameters is known *exactly* from the bilinear structure of the Limber projection—it is not

I think here people would like to know how to calculate $\Gamma(\mathcal{V})$ so define this explicitly

an approximation obtained by Taylor-expanding the Limber integral. The complete polynomial structure, including the quadratic term, is accessible from first principles for systematic assessment.

Second, the linear response matrix $\mathbf{\Gamma}(\boldsymbol{\vartheta})$ retains the full dependence on the cosmological and astrophysical parameters $\boldsymbol{\vartheta}$, inherited from the cosmology dependence of the basis tensor $\mathbf{B}_{uv}(\ell; \boldsymbol{\vartheta})$. In existing treatments, the effective covariance $\boldsymbol{\Sigma}$ is typically evaluated at a single fiducial cosmology and held fixed throughout the inference, an approximation that may suppress degeneracies between redshift distribution uncertainties and cosmological or astrophysical parameters. The tensorised approach, particularly when combined with the end-to-end emulation of Section 6.2, preserves the cosmology dependence of $\mathbf{\Gamma}$ at every point in parameter space, providing a principled baseline against which the fixed-cosmology approximation can be validated. In practice, this effect may be subdominant when the prior on the nuisance parameters is well constrained by external calibration data; the framework nevertheless provides the means to verify this assumption quantitatively.

Third, the entire marginalisation procedure operates on the precomputed basis tensor and does not require re-evaluation of the Limber projection, making it fully compatible with both emulated and directly evaluated pipelines. When combined with end-to-end emulation, analytic marginalisation further reduces the effective dimensionality of the sampling problem, enabling the MCMC chain to explore only the model parameters $\boldsymbol{\vartheta}$ while the redshift distribution nuisance parameters are integrated out at the likelihood level.

6.4 Future extensions

The preceding subsections demonstrated that the tensorised factorisation enables substantial computational acceleration (Section 6.1), a natural pathway to end-to-end emulation (Section 6.2), and exact analytic marginalisation over redshift distribution uncertainties (Section 6.3). These capabilities rely on the bilinear structure of the Limber projection (Equation 5.6) and the clean separation of cosmology-dependent coefficients from survey-specific distributions. We now outline three directions in which the framework can be extended, each of which preserves this factorisation and therefore inherits the same emulation and marginalisation infrastructure.

The most direct generalisation concerns non-flat geometries. Throughout this work, we have adopted the spatially flat assumption $f_K(\chi) = \chi$, which simplifies the lensing efficiency kernel and the multipole-wavenumber mapping. Although current CMB and baryon acoustic oscillation constraints restrict $|\Omega_K|$ to the per-mille level [23], a self-consistent treatment of spatial curvature is desirable for analyses that explore extended parameter spaces or marginalise over Ω_K . For non-zero curvature, the transverse comoving distance $f_K(\chi)$ (Equation 2.2) replaces χ in the denominator of the Limber integral (Equation 2.1) and in the lensing kernel (Equation 2.4). The piecewise-linear strategy extends naturally to this setting: within each redshift interval $[\chi_n, \chi_{n+1}]$, the ratio $f_K(\chi)/\chi$ varies slowly and can be linearised alongside the existing kernel and power-spectrum representations. The resulting integrands retain the rational-plus-logarithmic structure of the flat-sky case, with additional terms whose closed-form evaluation follows by the same methods used in the Appendix. Crucially, the linearised $f_K(\chi)$ depends only on cosmology and the redshift grid, so the factorisation into a cosmology-dependent basis tensor and survey-specific distributions is fully preserved, and only a modest increase in the number of analytic integral formulas per interval is expected.

Suggest removing this section and instead at the end of the summary section giving a shortened version as what else can be done.

Don't need to summarise preceding chapters unless you're writing the summary chapter.

The kernel decomposition of Section 4 is readily extensible to additional projected observables that share the same line-of-sight integral structure, as noted in Section 2. CMB lensing convergence provides the most immediate example: its radial kernel is structurally identical to $W_{\kappa}^a(\chi)$ (Equation 2.4) with the source distribution replaced by a delta function at the last-scattering surface, so it is already accommodated by the existing r_{κ} coefficients without any modification to the integral formulas. Cross-correlating CMB lensing with photometric galaxy clustering and cosmic shear extends the standard 3×2 pt framework to a 6×2 pt analysis, an approach increasingly adopted in current surveys [15, 17]; the tensorised formulation provides a natural and efficient infrastructure for such joint analyses. The integrated Sachs–Wolfe (ISW) effect introduces a radial kernel proportional to the time derivative of the gravitational potential, which can likewise be piecewise-linearised on the same redshift grid, adding a new set of basis-tensor components without modifying the existing machinery. Redshift-space distortions, which involve velocity-field kernels with a different radial weighting and angular-dependent terms, represent a further avenue for extension; their treatment requires additional care due to the derivative structure of the kernel and the anisotropy of the projection, but the underlying piecewise strategy remains applicable. Each of these extensions adds new kernel types to the existing framework while preserving the bilinear factorisation and its associated computational advantages.

A more challenging extension is the incorporation of exact angular power spectra beyond the Limber approximation. The Limber approximation is known to introduce per-cent-level errors at low multipoles ($\ell \lesssim 20$ –50), particularly for galaxy clustering and galaxy–galaxy lensing where the radial kernels have narrow support [69, 88]. The full (non-Limber) projection involves spherical Bessel functions $j_{\ell}(k\chi)$ whose rapid oscillations preclude the simple rational-function integrands that underpin the present framework. Two complementary strategies offer promising routes forward. The first builds directly on the piecewise formalism: within each redshift interval, the kernel is piecewise linear and the power spectrum is piecewise linear in χ , so the integrand takes the form of a low-order polynomial multiplied by a spherical Bessel function over a finite interval. Integrals of this type admit closed-form solutions in terms of generalised hypergeometric functions and spherical Bessel functions evaluated at the interval endpoints [72]; the same strategy can therefore be applied interval by interval, with the sparsity structure of the kernel coefficients exploited as in the Limber case. The second strategy employs FFTLog decompositions [86] to expand the power spectrum as a sum of power laws, $P(k) = \sum_m c_m k^{\nu_m}$. When combined with the piecewise-linear kernel representation, each term in the expansion yields integrals involving products of power laws and Bessel functions that are known analytically [70, 73–75]. This hybrid approach offers improved numerical stability at low multipoles where the Bessel oscillations are most problematic, while retaining the separability of the tensorised framework. In either case, the key factorisation property—that the cosmology-dependent coefficients and the survey-specific distributions enter as separate factors in a bilinear contraction—would be preserved, enabling the same emulation (Section 6.2) and analytic marginalisation (Section 6.3) strategies developed above. When combined with the non-flat geometry and additional-probe extensions, a beyond-Limber generalisation of LimberCloud would provide a unified, tensorised forward model for joint multi-probe analyses spanning photometric and spectroscopic surveys together with CMB observations, offering a scalable foundation for next-generation cosmological inference.

7 Summary and Conclusions

We have presented **LimberCloud**, a tensorised reformulation of the Limber projection in which every angular power spectrum of the 3×2 pt data vector is expressed as a bilinear contraction, $C_{uv}^{ab}(\ell) = \sum_{ij} B_{uv}^{ij}(\ell) \phi_i^a \phi_j^b$ (Equation 5.6). The basis tensor $B_{uv}^{ij}(\ell)$ encapsulates all dependence on cosmological and astrophysical parameters, while the discretised tomographic redshift distributions ϕ_i^a encode the survey-specific information. This factorisation is obtained by piecewise-linearising the radial kernels and the three-dimensional power spectrum on a discrete redshift grid, reducing each interval contribution to one of 21 structurally distinct closed-form integrals that populate the basis tensor without numerical quadrature.

Validated against **CCL** for the full LSST Y1 3×2 pt configuration—encompassing cosmic shear, galaxy–galaxy lensing, and galaxy clustering across all tomographic bin combinations—the framework achieves sub-per-cent accuracy over the multipole range $20 \leq \ell \leq 2000$, with typical residuals one to two orders of magnitude below the expected survey variance.

Benchmark timings demonstrate that the **JAX-GPU** implementation reduces the cost of the projection step to approximately 3 ms per evaluation for the full 3×2 pt data vector, more than three orders of magnitude faster than the conventional **CCL** pipeline. The cosmology stage—computing $\chi(z)$ and $P_{\delta\delta}(k, z)$ via **CAMB**—dominates the total wall-clock budget but is amenable to replacement by existing power-spectrum emulators. The tensorised structure further enables end-to-end emulation of the basis tensor itself: training a lightweight surrogate on $\sim 10^6$ parameter vectors is feasible within $\mathcal{O}(1)$ day on a modest GPU cluster and would reduce the per-evaluation cost to a neural-network forward pass plus the inexpensive projection contraction, yielding an expected speed-up of three to four orders of magnitude over the conventional approach.

The bilinear dependence of the angular power spectra on the redshift distributions yields an exact polynomial expansion of the data vector in the nuisance parameters (Equation 6.3), terminating at second order with no truncation error. Under a Gaussian prior, the linear-order contribution permits analytic marginalisation by completing the square, inflating the data covariance by the cosmology-dependent term $\mathbf{\Gamma} \mathbf{\Pi} \mathbf{\Gamma}^T$ (Equation 6.8). Higher-order corrections are accessible from first principles via Isserlis’ theorem and are expected to be small when photometric calibration priors are well constrained. This treatment avoids the finite-difference derivatives and fixed-cosmology approximations employed in existing analytic marginalisation schemes, providing a controlled framework for quantifying the adequacy of the linear-order approximation.

The framework extends naturally to non-flat geometries, to additional projected observables—including CMB lensing convergence, the integrated Sachs–Wolfe effect, and redshift-space distortions—and to beyond-Limber corrections via spherical-Bessel or FFTLog decompositions, all of which preserve the bilinear factorisation and its associated emulation and marginalisation infrastructure. Follow-up work will present end-to-end emulation within a differentiable inference pipeline and numerical validation of the analytic marginalisation formalism.

Acknowledgments

This study made extensive use of publicly available scientific software, including **NumPy** [95], **SciPy** [96], **Matplotlib** [97], and **JAX** [64]. We also gratefully acknowledge the computational resources provided by the Google Cloud Platform, whose GPU infrastructure was instrumental to this work.

Y.-H. Z. acknowledges support from the UK Research and Innovation (UKRI) Science and Technology Facilities Council (STFC) studentship, the Edinburgh–Leiden joint studentship awarded by the University of Edinburgh and Leiden University, and the STFC travel fund for UK participation in LSST (grant ST/X001334/1).

J. Z. acknowledges support from STFC funding for UK participation in LSST through grant ST/X001334/1.

NŠ is supported in part by the OpenUniverse effort, which is funded by NASA under JPL Contract Task 70-711320, 'Maximizing Science Exploitation of Simulated Cosmological Survey Data Across Surveys.'

M. A. receives support from the UK STFC through grant number ST/Y002652/1, as well as from the Royal Society via grant numbers RGSR2222268 and ICAR1231094.

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